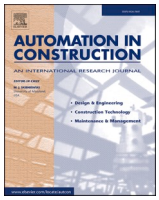
Automation in Construction 175 (2025) 106225



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Unified and interoperable digital twin models for emergency evacuation using building simulation identity cards

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ARTICLE INFO

Keywords:
Co-simulation
Digital twin (DT)
Building information modelling (BIM)
Model identity card (MIC)
BIM-based DT

ABSTRACT

Digital Twin (DT) technology is crucial for effective emergency evacuation planning in buildings, requiring specialized simulation models, from earthquake propagation to evacuation route planning. However, the lack of a unified vocabulary and standardized framework for integrating these models hinders seamless collaboration and limits holistic simulations. This fragmentation poses a significant challenge, as no single simulator can accurately capture the complex dynamics in emergency scenarios. To address this gap, this paper proposes the Building Simulation Identity Card (BSIC), a framework designed to standardize, characterize, and integrate the properties of building simulation models. BSIC facilitates the enhancement of traceability, reliability, and interoperability between DT systems in the building industry. This paper presents the application of BSIC in a case study focused on earthquake-induced evacuation in a multi-story office building. Findings demonstrate BSIC's efficacy in aligning discrete-event (DE) and continuous-time (CT) simulations, enabling synchronized data flows, accurate event handling, and minimized integration errors.

1. Introduction

The concept of the Digital Twin (DT) is becoming increasingly important in the Architecture, Engineering, Construction and Operations (AECO) sector as it enables real-time monitoring, simulation, and optimization of building performance throughout its lifecycle [1,2]. A DT represents a virtual replica of a physical entity, continuously evolving based on real-time data and operational history [3]. By leveraging DTs, stakeholders can predict how buildings will behave under various conditions, optimize energy use, and ensure structural safety and occupant well-being [4]. In building performance analysis, DT enables dynamic modelling of a building's physical structure and its operational behaviors, allowing for more accurate forecasting of energy consumption, evacuation analysis, and maintenance needs. Co-simulation is a critical component of DT which allows multiple simulation models to work together and represent the different facets of a building's performance [5].

Co-simulation integrates specialized simulations from different domains, each focusing on a specific aspect, such as structural integrity, energy consumption, or occupant behavior, making it an essential tool for the detailed and dynamic analysis required by DT [6]. Unlike monolithic simulations, where the entire system is modelled within one tool, co-simulation involves separate simulations for each component, coordinated and synchronized through coupling strategies and algorithms [7]. This method enables a more modular and flexible approach to system modelling, allowing mature tools to be utilized for each subsystem, though it introduces challenges in synchronization and result accuracy.

Simulation is a critical tool in modern building management, e.g., used to ensure both structural integrity and occupant safety during emergencies [8]. Disasters, whether natural or human-induced, present critical risks to life and property. Emergency evacuation, a fundamental safety strategy, systematically relocates individuals from hazardous zones to safety, minimizing potential harm [9]. Structural simulations predict how buildings respond to forces such as earthquakes, while evacuation simulations model human behavior during crises. However, addressing these aspects separately limits understanding complex interactions between building behavior and occupant response. Co-simulation, which integrates structural analysis with evacuation planning simulation, makes it possible to predict not only how a building will deform or collapse under stress, but also how people will respond to these changes.

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https://doi.org/10.1016/j.autcon.2025.106225
Received 3 December 2024; Received in revised form 8 April 2025; Accepted 18 April 2025
Available online 30 April 2025
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1.1. Problem statement

Co-simulation standards and techniques have become well-established in various mature technological industries such as computer science, aerospace, automotive, robotics, power systems, and telecommunications. These industries are primarily characterized by their reliance on cyber-physical systems [10], where physical processes are closely integrated with computational algorithms to form complex, dynamic systems. However, the products from the building industry (i.e., buildings) present unique challenges that hinder the direct application of co-simulation techniques from other domains of products (e.g., automobiles, airplanes, aerospace shuttles, etc.).

Unlike most products in other industries, a key distinction is that buildings are typically designed for extended operational lifespans, often spanning decades or even centuries, while being continuously occupied by people. We humans spend 90 % of our time in buildings [11], and during this time, buildings undergo changes in material conditions, energy consumption patterns, environmental impacts, and usage purposes. Co-simulations for the building industry must therefore consider long-term performance factors, such as material wear, energy efficiency, environmental impact, and even the adaptation of buildings to new functions over time. This makes co-simulation in the building industry fundamentally different from other domains where products have shorter, more predictable life cycles (e.g., automobiles, aircraft) and operate within well-defined CPS frameworks.

Moreover, the application of co-simulation in the building industry encounters several specific challenges:

- One of the most distinctive aspects of the building industry is its heavy emphasis on human and social factors [12]. While industries like automotive or aerospace focus primarily on the interaction between mechanical and cyber components, the building industry must account for the variability of human behavior, preferences, and societal interactions.
- While other industries rely on CPS for co-simulation, many buildings are not equipped with advanced sensors or automation systems [13]. Despite the absence of comprehensive cyber-physical integration, co-simulation is still necessary for various purposes, such as analyzing building evacuations or optimizing energy efficiency.
- Buildings are designed for a wide range of functions, e.g., schools, hospitals, offices, and residential spaces, each has its own design principles and user requirements [14]. This diversity complicates the development of a one-size-fits-all co-simulation approach, as each building type requires tailored models that accurately represent its unique structure and usage patterns, and performance requirements [15].
- Lastly, in complex building simulations, multiple simulation models are often developed by different domain experts, each focusing on specific aspects such as structural integrity, energy consumption, evacuation dynamics, or environmental impacts. These models, though highly specialized, are typically created using diverse tools, data formats, and protocols, leading to challenges in their integration during co-simulation [16]. The lack of a standardized way to communicate model specifications, assumptions, variables, and interactions poses a significant hurdle in achieving interoperability.

In addition to the above, there are also technical aspects and challenges in co-simulation that have to be taken into consideration when applying them in the building industry as follows:

- In the building industry, co-simulation often involves integrating simulation units that operate on different time scales. For example, a discrete-event (DE)-based model might simulate occupant evacuation behavior during emergencies, with events occurring at specific time points, while a Continuous-Time (CT)-based model simulates a building deformation, where changes happen continuously over time [17].
- Co-simulations in the building industry often involve the integration of models from various domains, such as structural analysis, HVAC performance, lighting, and human occupancy, each of which may use different tools, data structures, and units of measurement. Converting and mapping these data formats across different models can introduce inaccuracies, delays, and synchronization problems [18]. This is especially important in buildings, where long-term performance metrics like structural integrity must be closely monitored and accurately simulated over decades.
- Many simulation models used in the building industry operate as "black boxes", meaning they provide minimal insight into their internal workings [19]. These proprietary models, such as evacuation simulations, or structural analysis tools, often make critical assumptions or simplifications that are not visible to other models in the co-simulation. For example, assumptions about occupancy behavior or material ageing may not be explicitly communicated between models, leading to inconsistencies or incorrect interpretations of the simulation results.
- Co-simulating models with different computational demands can lead to performance bottlenecks. Some building simulations may require high-frequency updates and computations. In contrast, some simulations may remain idle for extended periods, resulting in an imbalance that increases overall simulation time or causes one model to lag behind another [20].
- When models from different domains interact incorrectly during a co-simulation, identifying the source of the error can be challenging [21]. For example, if a building evacuation model produces unexpected results, it may be difficult to determine whether the problem originated from the occupant behavior model, the fire spread model, or the structural integrity simulation. Without a standardized method for diagnosing where and why the synchronization, data exchange, or behavioral assumptions failed, the co-simulation process becomes error-prone and unreliable.
- Tightly coupled systems in building co-simulation often encounter time-step misalignment [22]. For instance, an occupant movement system might need high-frequency updates to reflect real-time changes in occupants' position, while a structural analysis might operate on much longer time steps. Misalignment between these time steps can lead to instability in the overall simulation, causing issues like incorrect energy performance predictions or structural failure.

In this paper, we will focus specifically on the challenge of integrating DE and CT simulations, exploring potential solutions to enhance synchronization, data exchange, and transparency in hybrid co-simulations in the building industry. In this framework, we address the following research question (RQ):

RQ: *How effectively does a framework for building simulation units facilitate novel integration of, and improve communication between, simulation models in a co-simulation process?*

By addressing the above research question, this paper aims to advance the applications of co-simulation in the future development of DT technology in the building industry by developing a framework that helps promote consistency in data exchange, minimize integration errors, and foster cross-domain collaboration, ultimately improving the accuracy, traceability, and reliability of simulation outcomes.

1.2. Research methodology

The methodology for this paper is designed to address the research question (RQ) through interconnected phases: a literature review, the development of the Building Simulation Identity Card (BSIC) framework, and its application in a case study. Each phase aligns with the research objectives and tackles specific challenges in co-simulation within the building industry. The paper starts with a literature review

aimed at understanding existing standards for co-simulation in various industries and exploring related work in the building sector. This review highlights challenges and successful approaches in integrating multiple simulation models while identifying gaps in long-term performance, human dynamics, and building functions. The gaps explored particularly focus on barriers to simulation model integration (addressing the first part of the *RQ*), and challenges in communication between simulation models (addressing the second part of the *RQ*). These insights support the development of the BSIC framework as introduced in our previous paper [23], which aims to enhance interoperability and communication among simulation models in the building industry.

The next step involves the development of the BSIC framework itself. The BSIC standardizes the key attributes necessary for integrating simulation models, such as time synchronization, data exchange formats, and event trigger mechanisms. Each attribute of the BSIC framework is validated through a case study to ensure its direct contribution to enhancing interoperability and communication between simulation models, thereby aligning with the core objective of this paper.

The final phase involves the application of the BSIC framework in a case study focused on co-simulating structural integrity and occupant evacuation during an emergency scenario in a building. The case study allows for testing the effectiveness of BSIC in ensuring that DE, such as structural failures, are accurately synchronized with CT processes, such as continuous occupant movement, to produce reliable and precise simulation outcomes.

The case study is designed to evaluate the BSIC framework's ability to address the *RQ* by simulating a real-world emergency scenario in a five-story office building. The aim of the case study is to provide new evidence that the BSIC framework facilitates the seamless integration of diverse simulation models by standardizing data exchange formats, synchronizing time steps, and ensuring accurate event propagation between DE and CT models.

The results of the case study demonstrate that the BSIC framework enables communication between semantically and technically diverse simulation models, i.e. the structural failure simulator that delivers output *events* (failures) based on structural integrity and shear forces, and an occupant simulator that delivers *continuous* positional updates during an evacuation, where the structural failure events have a direct impact on occupant behavior. By synchronizing the structural failures with occupant movement, while minimizing integration errors, the results of the case study demonstrate a novel integration of the two distinct simulation models that would otherwise not be combined within a single unified (co-) simulation. Thus, these findings directly address the *RQ* by providing empirical evidence of the framework's ability to improve co-simulation processes, ensuring both theoretical and practical applicability in the building industry.

The remainder of this paper is structured as follows: Section 2 provides the necessary background by introducing the concept of DT and exploring relevant standards and frameworks that support co-simulation across different domains. Section 3 reviews related work, critically examining existing approaches and methodologies in co-simulation and evacuation analysis in the building industry. Section 4 provides the BSIC framework, detailing its architecture and the mechanisms through which it facilitates model integration. Section 5 demonstrates the practical application of the BSIC through a detailed case study, whereas section 6 discusses the broader implications of the framework, particularly in terms of its contribution to improving interdisciplinary collaboration and simulation accuracy. Finally, Section 7 provides concluding remarks, summarizing key insights from the study, acknowledging limitations, and proposing avenues for future research. The research structure of the paper is highlighted in Fig. 1.

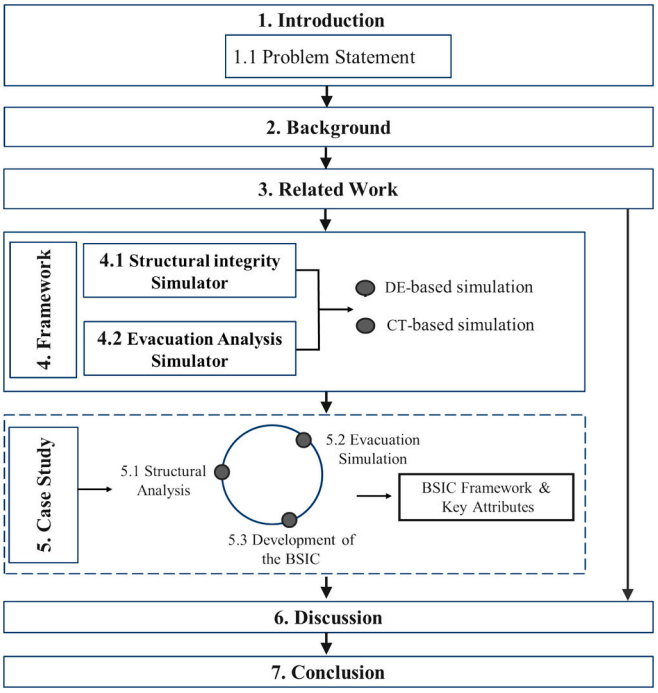


Fig. 1. Overview of research structure.

2. Background

2.1. Standards and framework to support the process of co-simulation

Various methodologies and frameworks have been proposed to facilitate the interaction of simulation models, aiming to enhance the communication between simulation models used in co-simulation. For instance, the Functional Mock-up Interface (FMI) standard has been developed as an open format for model exchange and co-simulation [22]. However, the FMI standard has identified several challenges in the domain of co-simulation, particularly in managing event synchronization and input-output dependencies across multiple simulation models, especially when operating with different clocks [24]. Presents a comprehensive comparison of five key frameworks, i.e. SSP, MIC, SMP, MoSSEC and HLA, employed in engineering and simulation processes. The System Structure and Parameterization (SSP) standard provides a structured, tool-independent framework for defining, exchanging, and packaging complex systems made up of multiple simulation components. It standardizes the Parameters such as material properties, environmental conditions, and control inputs using SSV files. It enables seamless integration of models from different domains, enhancing modularity, scalability, and reusability in co-simulation environments. SSP is widely applied in industries like aerospace, where it facilitates the integration of CAD models with system simulations for tasks such as aircraft development [25]. Despite its advantages, ensuring interoperability across tools with different levels of SSP support can be challenging, particularly as the standard is still maturing [26].

The Model Identity Card (MIC) is a standardized framework designed to document and manage simulation models in collaborative, multidisciplinary design environments. It facilitates knowledge sharing between system architects and model providers by specifying input/output parameters, behavior, and interfaces, ensuring consistency and accuracy in model integration. For instance, in vehicle crash simulations, MIC ensures all stakeholders have a clear understanding of how structural deformation and occupant safety models interact. MIC is particularly useful in complex systems development, such as automotive and aerospace, where it aids in integrating diverse models and improving early-stage verification and validation. However, it has some challenges,

including ensuring interface consistency and managing the complexity of integration across different domains [27].

Simulation Model Portability (SMP) is a standard developed by the European Space Agency (ESA) to ensure the reusability and interoperability of simulation models across different platforms. Its primary purpose is to enable seamless transfer and reuse of models without significant reconfiguration [28]. SMP finds applications in various domains, including radar systems, space exploration, and satellite simulations, where it supports both structural and behavioral modelling [29,30]. However, it also highlights some challenges, including difficulties in fully representing complex dynamic behaviors and achieving model interoperability when integrating models developed using different standards, such as synchronous data flow and domain-specific languages [29,30].

MoSSEC (Modelling and Simulation information in a Collaborative Systems Engineering Context), formalized as ISO 10303-243, is a standard designed to manage and exchange systems engineering data across multiple stakeholders in collaborative projects. It provides a structured format for sharing key metadata associated with models, simulations, and system decisions, ensuring traceability and interoperability [31]. In the aerospace industry, MoSSEC is critical for long-term archiving and retrieval, allowing teams to capture the rationale, context, and intent behind models, essential for certification, reuse, and regulatory compliance [32]. However, MoSSEC faces challenges, including incomplete object associations, unclear handling of similar component types, and ensuring tool interoperability with consistent metadata. Additionally, scalability and performance issues arise in managing large-scale, complex systems [33].

High-Level Architecture (HLA) is a standardized framework for distributed simulation developed to provide a common architecture for modelling and simulation. It is designed to promote interoperability, reusability, and scalability in complex simulations [34]. Originally, HLA was developed for military purposes that facilitate communication between heterogeneous systems (called federates) within a simulation environment. Data exchanges occur through publish-subscribe patterns, where federates publish events and other federates subscribe to updates. HLA's applications span various fields, such as disaster management, smart grids, infrastructure resilience, and construction project monitoring, enabling efficient co-simulation across different domains [35,36]. HLA's time management services ensure that federates with varying time steps remain synchronized. However, HLA faces several challenges, e.g. its integration across diverse federates can be complex, especially in synchronizing time management and communication between systems with different time-steps or event frequencies [35]. Table 1 presents a comparative analysis of the aforementioned frameworks in facilitating interaction among various simulation models.

3. Related work

To ensure a comprehensive review of relevant literature, we utilized the Web of Science database as the primary source for identifying scholarly articles related to co-simulation frameworks, and emergency evacuation modelling in the building industry. The search was conducted using a combination of keywords, including co-simulation in buildings, emergency evacuation modelling, earthquake analysis, interoperability in simulation and BIM-based simulation. The inclusion criteria focused on the studies that addressed interdisciplinary simulation challenges, synchronization methods, and data exchange protocols. Additionally, articles that specifically discussed the integration of DE and CT simulation paradigms were prioritized.

3.1. Co-simulation in building industry related studies

Co-simulation in buildings has emerged as a powerful method for integrating multiple simulation domains, enabling a more comprehensive analysis of building performance under varying conditions [37,38]. Various studies have applied co-simulation techniques to address specific challenges in building design, operation, and maintenance. A study by Yeung et al. [38] proposed an open BIM-based co-simulation architecture that integrates Building Energy Simulation and Life Cycle Assessment (LCA). The architecture combines EnergyPlus and Brightway2, enabling dynamic, time-differentiated results for evaluating energy use and environmental impacts. While this approach provides a foundation for dynamic integration, it is reliant on proprietary BIM tools, which raise accessibility concerns and limits interoperability with non-BIM systems. Furthermore, the study does not address the scalability of this framework for use in large building stocks or urban-scale simulations. Similarly, a study by Cellura et al. [39] presents an approach that integrates building simulation and LCA through the development of a TRNSYS component capable of performing LCA studies within the same simulation environment. This approach allows for a comprehensive analysis of both the energy and environmental impact analysis, focusing on Net Zero Energy Buildings. However, the work is limited by the need for manual data integration for specific applications. Additionally, the study does not adequately explore how the integration could handle the temporal variability of energy use patterns across different climate zones.

Yang et al. [35] developed an overarching HLA-based co-simulation framework for interdependent infrastructure resilience management. It facilitates the dynamic modelling of time-varying resilience behavior and promotes collaborative decision-making among stakeholders through seamless interoperability and message exchange. While the proposed overarching HLA-based co-simulation framework offers

Table 1
Comparative analysis of SSP, MIC, SMP, MoSSEC, HLA modelling and simulation frameworks.

Aspect	SSP	MIC	SMP	MoSSEC	HLA
Focus	System structure and parameter exchange	Standardized simulation model metadata	Model portability and reusability	Collaborative alignment of Requirements, Functional, Logical, and Physical domains	Distributed simulation and optimization
Objective	Enable modular, multi-domain engineering	Reduce inconsistencies and improve collaboration	Improve portability of dynamic simulation models	Enhance traceability and integration across domains	Interoperability in distributed systems
Strengths	Open standard, XML-based, flexible scalability	Consistent, traceable, and early-stage error detection	Integrates dynamic and static behavior models	Supports iterative, collaborative design processes	Modular and flexible real-time frameworks
Applications	Aerospace, complex multi-domain systems	Automotive, aerospace, and multidisciplinary design	Radar systems, reusable simulation components	Aerospace, high-value design, and engineering	Distributed, real-time systems and optimization
Technical Details	Defines modular architectures via SSD files and parameterizes simulations with SSV files for consistency.	Documents metadata and define input-output interfaces, and validates integration early to reduce errors.	Ensures portability and standardization of reusable simulation components.	Shares system models and simulation results collaboratively across enterprises.	Facilitates event-driven synchronization and communication via RTI for hybrid simulations.
Challenges	Tool compatibility and complexity	Consensus on vocabulary and standards	Limited semantic support for complex models	High overhead for setup and process alignment	Synchronization and communication overhead

several advantages, it poses challenges for stakeholders due to the complexity of integrating domain-specific simulators, requiring modelers to acquire cross-domain knowledge and redesign interfaces for different simulators. This limits its usability for practitioners or in scenarios requiring rapid deployment.

Benzaama et al. [40] used a co-simulation approach integrating TRNSYS with FLUENT, simulating air temperature, surface conditions, and the dynamic behavior of indoor air in building spaces. The study demonstrates the effectiveness of a combined geothermal system and earth-air heat exchanger for improving thermal behavior in buildings across various Algerian climates, particularly in Oran, Tamanrasset, and Bechar. While the study is contextually relevant, its findings are highly location-specific, limiting generalizability to other regions with different climatic conditions or building types.

Huo et al. [41] utilized co-simulation techniques to explore energy efficiency and emission reduction strategies in buildings, but with a different focus and methodology. They combined a system dynamics model with the Long-range Energy Alternatives Planning System model, embedding a bottom-up building end-use energy model into the system dynamic framework to explore carbon emissions. This integration allowed them to capture the dynamic feedback between macroeconomic factors and micro-level building energy use. Additionally, by incorporating Monte Carlo simulations, the authors accounted for uncertainties in future predictions, revealing that the carbon emissions for the urban residential building sector in Chongqing, China, are projected to peak at 22.8 Mt. CO_2 around 2042 ($\pm$3.4 years). While the study provides robust insights into emission trends and sensitivity analysis, it lacks granularity in analyzing specific end-use interactions, such as the interplay between thermal systems and occupant behaviors.

Taveres et al. [42] in their paper focused on co-simulation as a technique to predict performance in advanced building envelopes, such as adaptive facades. These facades often use complex systems like dynamic insulation or switchable glazing, which respond to environmental conditions. Co-simulation allows simultaneous data exchange between different simulation models, enabling more precise performance predictions across multiple physical domains (e.g., thermal, lighting, moisture). While the study highlights the potential of co-simulation for advanced systems like dynamic insulation and switchable glazing, it lacks a detailed discussion on the computational resources required and the feasibility of implementing such systems in cost-sensitive projects.

Similarly, a study by Peng et al. [43] focuses on optimizing building performance by highlighting the ability of hybrid control systems to integrate natural ventilation and HVAC (Heating, ventilation, and air conditioning) for real-time optimization in mixed-mode buildings, improving energy efficiency and comfort. The paper discusses energy savings potentials ranging from 8 % to 78 %, depending on climate conditions, with reductions in cooling energy consumption of up to 54.4 % in warm climates. However, the study does not address the practical challenges of integrating advanced control strategies, such as model predictive control in existing buildings, particularly those without pre-installed IoT infrastructure. This gap limits its potential for retrofitting applications, a critical area for achieving widespread energy efficiency improvements.

Additionally, a study by Frank et al. [44] developed a co-simulation architecture that integrates EnergyPlus® with a new Modelica library to model electrical distribution systems with much finer detail than previously possible. This combination enables simultaneous simulations of building energy and electrical systems, allowing researchers to more precisely assess interactions between HVAC loads and electrical efficiency. The results showed that improving electrical distribution efficiency could significantly reduce cooling loads, particularly in high-performance buildings. Furthermore, a study by Barbierato et al. [45] discussed the scalability of co-simulation frameworks like Mosaik and HELICS for multi-energy systems. These frameworks enable the integration of different simulators, managing issues such as time synchronization and data transfer between systems, which makes them particularly valuable for large-scale building and energy grid simulations.

3.2. Structural and evacuation dynamics in emergency scenario related studies

Several studies have explored emergency evacuation simulations in buildings. For instance, a study by Forssberg et al. [46] focused on improving evacuation design reliability by analyzing the variability in pre-movement times during evacuations. The researchers used video recordings to capture a total of 2486 individual pre-movement times and statistically analyzed their variability. The results from the study offer fire safety professionals enhanced predictive capabilities for evacuation times. However, the study also faced challenges, including difficulties in differentiating between the recognition and response phases, limitations in video coverage, and variations in alarm responses influenced by social and situational factors. In parallel, Feng et al. [47] presents an emergency fire escape path planning model that utilizes an enhanced Deep Deterministic Policy Gradient algorithm. This model integrates fire-related data with reinforcement learning to dynamically generate optimal evacuation routes while considering fire hazards such as smoke and heat. The algorithm used is improved through the incorporation of intrinsic motivation, hindsight experience replay, and hyperparameter optimization via the Beluga Whale Optimization algorithm, leading to efficient training and effective pathfinding capabilities. Although the study demonstrates technical sophistication, it does not sufficiently address the computational resources required for real-time applications, which is a practical barrier in emergency scenarios.

Additionally, Kirby et al. [48] used AnyLogic simulation software to optimize the operations of a Regional Hub Reception Centre (RHRC), which is designed to process 20,000 disaster evacuees over a seven-day period. While the study focuses on critical disaster response metrics such as throughput time and space requirements, it does not explore the implications of scaling this framework to larger urban centers or handling simultaneous disasters, which are essential considerations for disaster preparedness in densely populated areas.

The author of the paper [49] proposed a network-based simulation model for simulating earthquake evacuation processes in multi-story buildings by combining pedestrian dynamics, probabilistic structural damage assessment, and structural random vibration analysis. The model integrates pedestrian behavior and structural collapse during an earthquake to simulate evacuation scenarios. However, it also highlighted a few challenges, including the simplification of complex human behaviors and refining the analysis of structural damage for more accurate results. Similarly, a study by Hassanpour et al. [50] also highlight the importance of combining structural and behavioral modelling for more accurate evacuation predictions. They developed a post-earthquake evacuation model utilizing an Agent-Based Building Earthquake Evacuation Simulation (AB^2E^2S). This model integrates agent-based modelling techniques with BIM, focusing on the interaction between building occupants and the environment during evacuations. The simulation assesses the impact of architectural design, non-structural damage, and human behavior on evacuation efficiency. However, the model simplifies human behavior and structural dynamics, which could lead to inaccuracies in predicting real-world evacuation outcomes, particularly in scenarios involving highly variable human responses.

Spearpoint et al. [51] reviewed various scientific studies and employed two primary simulation tools, Pathfinder and Evacuationz, to model high-rise residential evacuations. It stress-tests building designs by comparing the performance of these tools under different conditions, such as stair widths, building heights, and agent behavior, to assess the potential evacuation outcomes and identify optimal design strategies. Similarly, Chu et al. [52] in their study propose a framework for optimizing emergency evacuation management in urban residential communities. Their approach integrates both indoor and outdoor evacuation simulations while factoring in family behaviors and spatial data. The

model identifies critical evacuation bottlenecks through multi-scenario analysis like crowded stairways and congested roads. The study highlighted the challenges that primarily involve collecting ultra-detailed spatial and resident data, which limits its scalability to small communities.

4. Framework for the practical application of BSIC

Understanding how multiple simulations function within the DT framework is essential to understanding the BSIC's development process. Fig. 2 illustrates an abstract demonstration of the system where real-time data from building sensors is integrated with a digital model to manage multiple simulations in parallel. Various sensors installed within the building gather data on environmental and structural conditions, while alarms and changes within the building are activated based on the outcomes of the simulations. These outcomes involve "what-if analysis," where potential scenarios—such as structural failures, evacuation bottlenecks, or environmental hazards—are evaluated within the simulation environment to predict the impact of such events.

This data is transmitted to a central gateway or coordinator responsible for managing and synchronizing various simulations (e.g., Structural analysis, Evacuation analysis, etc.) within the DT environment. At the core of this architecture are the layered components of DT, as represented in Fig. 3. The data acquisition layer serves as the foundation, gathering real-time information from IoT devices and sensors embedded in the physical environment. This data is then securely transmitted through the transmission layer, leveraging robust communication technologies such as Wi-Fi and 5G to ensure seamless connectivity. The digital modelling layer processes this data to construct dynamic simulations, accurately replicating the physical and functional aspects of the building. These simulations are further refined and integrated in the data-model integration layer, enabling sophisticated analytics, predictive modelling, and AI-driven insights. The service layer builds on these capabilities, providing stakeholders with decision-making tools and interactive visualization, fostering operational efficiency, informed decision-making, and sustainability.

Each simulation model will have its BSIC, enabling seamless communication and data exchange between simulation models to enhance coordinated decision-making and response.

BSIC concept is developed to address the challenges of integrating and orchestrating specialized simulators within a collaborative simulation framework. This approach draws inspiration from key technological initiatives, such as the MIC in the automotive industry, which characterizes simulators to improve interoperability. BSIC attributes were refined and expanded through iterative testing, using data flows between structural analysis and evacuation simulators. The case study enabled the identification and integration of additional attributes.

To address occupant safety challenges in buildings, a multidisciplinary design and engineering team typically creates a comprehensive digital model of the structure, often utilizing BIM-based software such as Revit. A structural engineer then analyses the building for potential failures and risks under earthquake loading [54], using the BIM model in a structural integrity simulator. The earthquake loading data, sourced from the PEER Ground Motion Database [55], must be integrated into an evacuation planning simulator. This integration involves transferring data related to building occupants, outputs from the structural integrity analysis, and available evacuation routes. The ultimate goal of these simulations is to develop an evacuation strategy in response to seismic events.

However, neither the structural integrity simulator nor the evacuation planning simulator can independently execute the full scenario, necessitating a co-simulation to achieve accurate and reliable results. A significant challenge in this process is the lack of a common vocabulary or standardized framework for integrating different simulators, increasing the risk of error in the process of co-simulation.

The integration of outcomes derived from the seismic analysis simulator into the evacuation planning simulator has been executed manually. As shown in Fig. 4, the framework for co-simulations is applied to emergency evacuation planning from buildings. The evacuation planning engineer uses the output from the structural integrity simulator, gathered by BSIC, as input for the emergency evacuation simulator.

4.1. *Sequence diagram for COE-driven Co-simulation process*

A master algorithm is utilized by a Co-simulation Orchestration Engine (COE), such as the Jacobi algorithm (for detail refer to Appendix A), which performs a simulation step by initiating the advancement of all FMUs in time, retrieving the necessary outputs, and setting the necessary inputs. The simulation follows a series of iterative steps until a termination condition is met, typically determined by the maximum timestep specified by the user. The procedural sequence can be described as follows: The COE initiates the simulation cycle by dispatching inputs to all FMUs. Subsequently, the COE retrieves the next time step from each FMU. Advancing the simulation, the COE progresses the state of each FMU's simulation to the minimum (earliest) time reported by the respective units. Finally, the COE consolidates the output data generated by all FMUs.

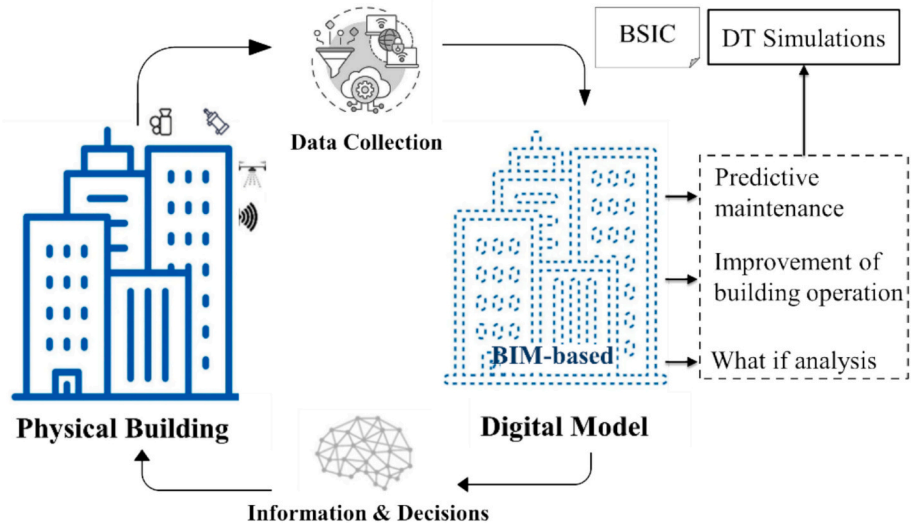


Fig. 2. Integration of physical building and digital model for enhanced simulation with the help of BSIC.

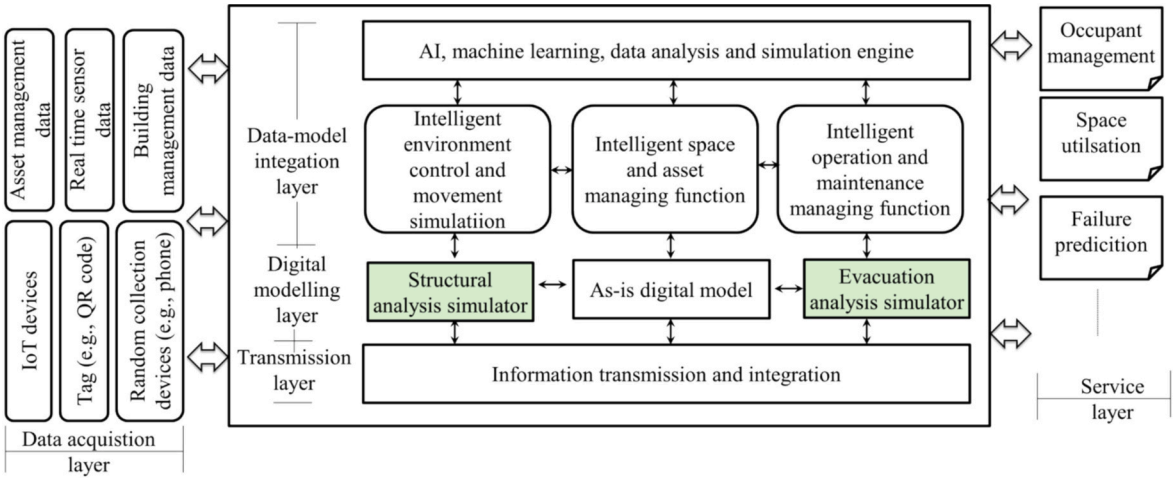


Fig. 3. System architecture of DT development in buildings (reproduced from [53]).

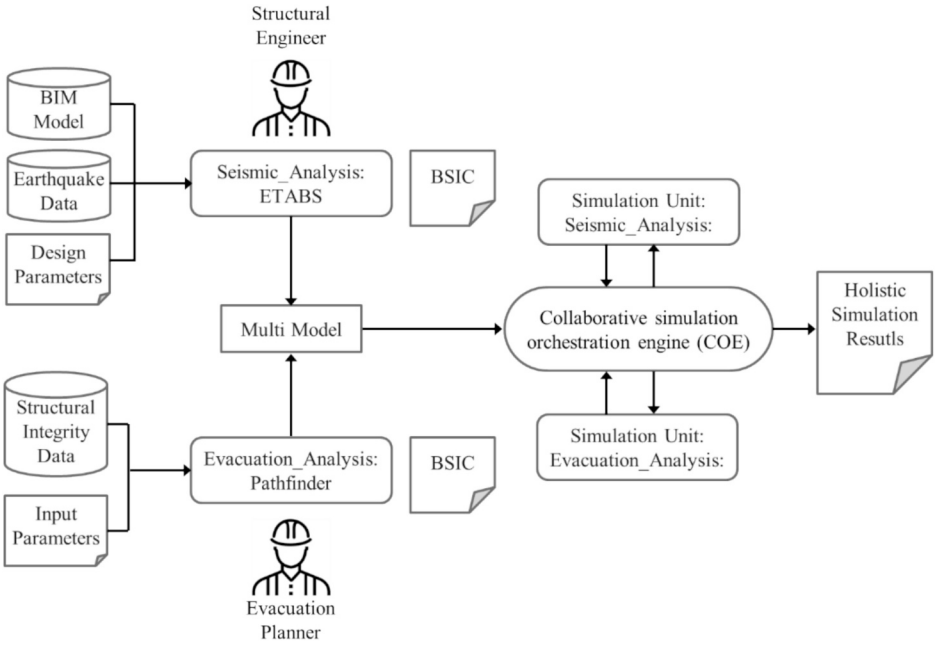


Fig. 4. Framework for co-simulations for emergency evacuation from buildings. For detailed attributes of BSIC refer to Fig. 16.

Fig. 5 presents a sequence diagram [56], illustrating the interactions between different components of the case study system (ETABS and Evacuation analysis simulator) in a step-by-step manner over time. The solid vertical lines indicate the progression of time for each entity. The horizontal arrows show the communication or data exchange between entities (e.g., parameter updates, status sharing), and the dotted lines signify the return messages. This diagram illustrates the iterative and synchronized operations required to effectively simulate structural and evacuation dynamics by integrating seismic analysis outcomes into evacuation planning simulations within a manually conducted co-simulation framework. The COE manages the interaction between different FMUs, each representing independent simulators. In each simulation iteration, the COE initiates the sequence by dispatching necessary inputs to all FMUs, retrieving the upcoming milestone time from each FMU, and advancing the simulation to the earliest reported time. Subsequently, the COE collects the output data from all FMUs and uses this information to proceed to the next iteration.

The iterative simulation process described follows a time-stepping approach, where the COE ensures synchronization between various simulators, such as those modelling structural behavior (e.g., seismic analysis using ETABS) and evacuation dynamics. For instance, as illustrated, in the first Jacobi iteration, the simulation is advanced to a $t = 14$ s milestone, where structural data (e.g., beam collapse) is passed to the evacuation simulation. The system continues iterating until it reaches the maximum time step during which it consolidates input and output from all simulators.

Fig. 6 demonstrates that the ETABS simulator operates two distinct types of event-based simulations: DE and CT simulations. The DE-based simulation focuses on applied loads at specific time milestones, like at 14 s, where it calculates the maximum load applied to the structure. In contrast, the CT-based simulation tracks the continuous deformation of the structure over time.

To accurately model both the structural integrity of buildings and evacuation scenarios under dynamic conditions, we employ a co-simulation framework that integrates both DE-based and CT-based simulations. This approach is crucial for capturing the complex interactions between structural deformations due to loads (e.g., earthquakes) and the movement of occupants during evacuation.

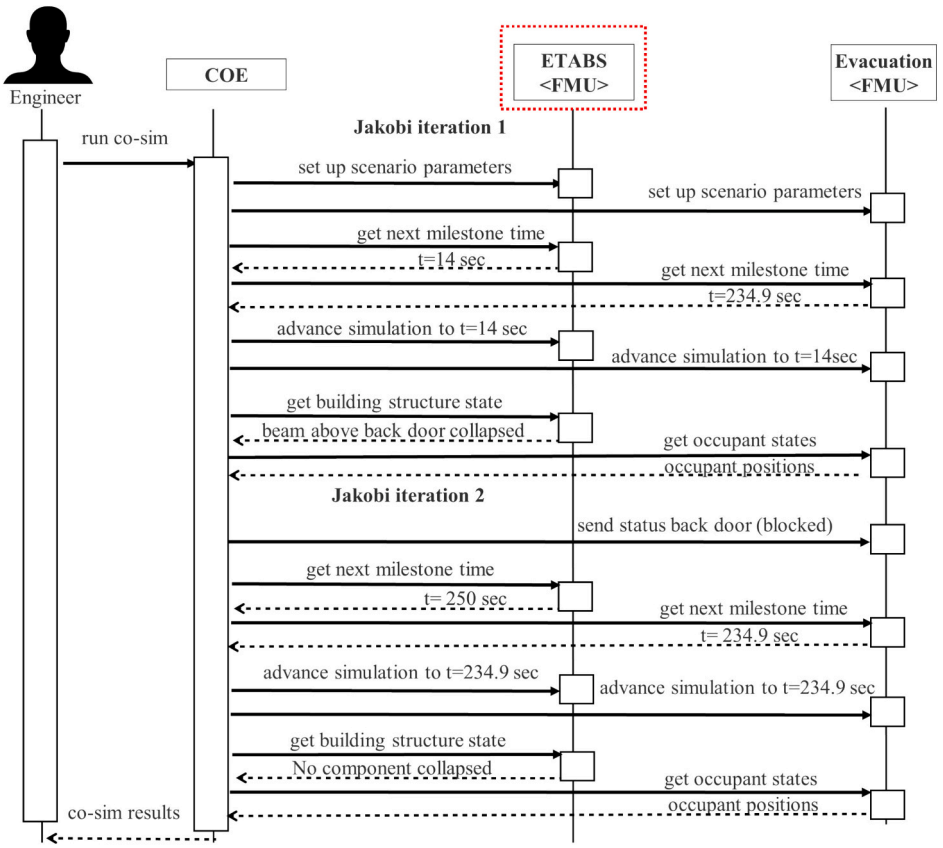


Fig. 5. Sequence diagram for co-simulation process in emergency evacuation planning. The red box is for the ETABS simulator, which is explained Fig. 6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

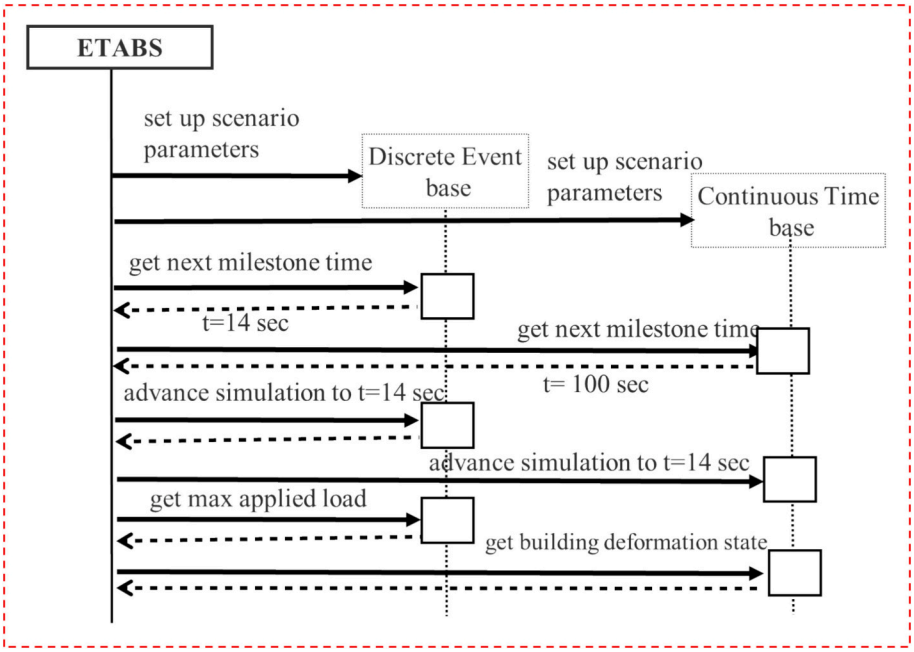


Fig. 6. Sequence diagram to explain the DE and CT simulations for ETABS simulator.

4.2. Structural integrity simulator

The structural integrity simulator operates within a hybrid co-simulation framework in which the structure's deformation is modelled continuously, and the application or removal of loads, such as those resulting from an earthquake, are treated as DEs.

4.2.1. DE-based simulation

The DE part handles events such as the application or removal of loads, which cause instantaneous changes in the state of the structure.

This relationship is defined by the transition function in expression (1), which describes the system transition from its current state to a new state when an external event occurs.

$$X_{ext}(x, e, t) \rightarrow \text{x'} \tag{1}$$

Where:

- X_{ext}: External transition function
- x: The current state of the system
- e: External events (e.g., earthquake load)
- t : Elapsed time since the last event
- x' :The new state after the external event

4.2.2. CT-based simulation

For the continuous part, we use a state-space representation to model the deformation of the structure. This representation describes the changes in the structure's state over time based on its current condition and any external inputs. The state Eq. (2) describes the system's evolution, such as changes in the deformation of the structure over time. Meanwhile, the output Eq. (3) relates the internal state to the observed output, such as measurable responses of the structure, including stress, strain, velocity, or acceleration at specific points.

$$x(t)' = \mathbf{A}\, x(t) + \mathbf{B}\, u(t) \tag{2}$$

$$y(t) = \mathbf{C}\, x(t) + \mathbf{D}\, u(t) \tag{3}$$

Where:

- $x(t)$:State vector representing the structure's deformation ($t \in \mathbb{R}$), where $t \geq 0$ and $\mathbb{R}$ is a set of real numbers
- $y(t)$:Output vector represents measurable quantities (e.g., stress, strain)
- $x(t)'$:Time derivative of the state vector (change in deformation of the structure over time)
- $u(t)$: Input vector, representing external forces (e.g., earthquake loading)
- **A** :State matrix captures the system's inherent properties (such as stiffness and damping effects)
- **B** : Input matrix, which scales the impact of external influences on the system's state
- **C** : Output matrix (determines how the internal state $x(t)$ relates to outputs $y(t)$ e.g., displacement, stress)
- **D** : Direct transmission matrix (representing how external influences affect the measurable outputs)

Fig. 7 illustrates the interaction between DE-based and CT-based simulations in structural analysis. The DE-based simulation focuses on the initial state of the structure, the intensity of the applied load, and the specific timing of that load. It calculates the load's effect at discrete moments. In contrast, the CT-based simulation tracks the structural response over time, considering external influences on produced stress and strain. It evaluates the speed of deformation, factoring in the structure's initial state and ongoing conditions. Together, these simulations provide a detailed analysis of both load application and the continuous deformation process under external forces.

4.3. Evacuation analysis simulator

The evacuation analysis simulator also follows a hybrid co-simulation approach. The continuous time-based simulation captures the movement of occupants, while DEs account for changes in the environment, such as temporary obstructions or barriers.

4.3.1. DE-based simulation

The model accounts for specific actions that occur at particular moments, such as temporary obstructions. These discrete actions are governed by the transition function, defined in expression (4), which determines how the state changes in response to these events.

$$X_{ext}(x, e, z, t) \rightarrow \text{x'} \tag{4}$$

Where:

- X_{ext}: External transition function
- x: The current positions and speed of people
- e: An event, like creation of an obstructions (collapse of beam)
- z: Building condition event from the Structural Simulator
- t : Elapsed time since the last event
- x' :The new positions and speed after the external event

4.3.2. CT-based simulation

For the continuous part, we use a state-space model to represent the movement of people. The state-space model formalizes and communicates the continuous nature of evacuation dynamics without directly modelling complex individual behaviors, which are handled in evacuation simulations. The state Eq. (5) represents the rate of change in positions and velocities over time, while the output Eq. (6) provides measurable quantities, such as the number of people in a specific zone.

$$x(t)' = f(x(t), u(t), z(t)) \tag{5}$$

$$y(t) = g(x(t), u(t), z(t)) \tag{6}$$

Where:

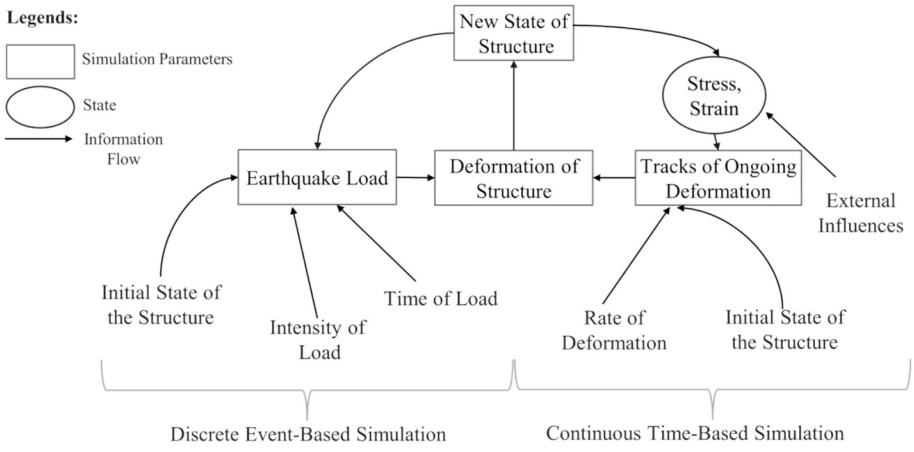


Fig. 7. ETABS logic diagram of the simulation process.

- $y(t)$:Output representing measurable quantities, e.g., the number of people in a specific zone ($t \in \mathbb{R}$) where $t \geq 0$
- $x(t)$' :Time derivative of the state vector representing the rate of change of positions and velocities
- $f(x(t), u(t), z(t))$:A dynamic function that describes how people's positions and velocities evolve in response to:
  - o $x(t)$:State vector representing positions and velocities of people
  - o $u(t)$: Input vector, representing external influences such as obstacles
  - o $z(t)$: Building condition vector, representing real-time data from the structural simulator

Fig. 8 illustrates the interaction between DE and CT simulations in evacuation analysis. The DE-based simulation focuses on specific events that occur at specific moments, such as collapse of beam. It is based on the information provided by structural analysis simulation and the updated structural state after the application of load. These DEs influence the flow and movement of people within the structure. In contrast, the CT-based simulation tracks the continuous movement of individuals over time, monitoring their positions and velocities as they respond to changing conditions and obstacles.

5. Case study

To address the RQ, this paper employs a case study focused on emergency evacuation scenarios. The RQ aims to enhance communication, synchronization, and interoperability between simulation models within co-simulation processes. Emergency evacuation scenarios were selected as they present inherently complex and interdisciplinary challenges. These scenarios necessitate the integration of DE simulations, such as structural component failure during seismic events, with continuous-time CT simulations, such as occupant movement during evacuation. The synchronization and data exchange requirements in such scenarios are highly intricate, making them an optimal context for validating the proposed BSIC framework. By applying the BSIC framework in this case study, the research demonstrates its efficacy in aligning DE and CT simulations, ensuring accurate and reliable simulation outcomes, thereby addressing the core objectives of the research question.

The building under examination as illustrated in Fig. 9 is a five-story building located in Islamabad, Pakistan, and serves as an office facility with a seating capacity for 170 individuals. The building was completed in 2018. Its reinforced concrete structural system, designed to withstand vertical and lateral loads, ensures long-term stability and makes it an ideal choice for ETABS analysis, especially given Islamabad's seismic conditions. The facade is constructed with glass. The configuration of the façade not only enhances the aesthetic value of the structure but also contributes to its energy efficiency by optimizing the use of natural light. The front of the building features two primary doors for regular entry and exit. Additionally, there is a rear door designated for service and maintenance access. For safety purposes, the building is also equipped with an emergency exit door to facilitate evacuation in case of emergencies (refer to Fig. 8(b)). ETABS and Pathfinder were selected for their specialized capabilities in structural analysis and evacuation modelling, providing a comprehensive evaluation of both performance and safety.

5.1. Structural analysis

A comprehensive structural analysis of the building is conducted, revealing a fundamental time period [57] of 0.96 s, i.e. the time taken by the building to complete one cycle of vibration under applied load. ETABS software [58] is used for an analysis of structural stability and earthquake impact [59]. ETABS has been widely utilized in seismic analysis and structural design for multi-story buildings, demonstrating its capability to model dynamic structural behaviors and accurately predict deformations and failures under seismic loads [60]. The time history of the 1989 Loma Prieta earthquake [55] served as a reference for this analysis.

Fig. 10 shows the structural model from ETABS used to design the case under consideration. The structural analysis reveals the maximum stresses experienced by structural elements at $t = 14$ s, with corresponding shear and moment recorded. During this critical instance, the shear and moment data suggest a potential beam failure above the back door. This is because the design shear and moment capacity of the beam is less than the shear and moment caused by the applied loads. Subsequently, these findings were transcribed into the BSIC, forming the basis for input in an emergency evacuation analysis.

Fig. 11 represents the beam above the back door at Story GF (12 × 18) under the applied load combination. According to the Building Code of Pakistan (BCP) [61], the allowable shear capacity ($V_{allowable}$) for this beam can be calculated using the formula in Eq. (7):

$$V_{allowable} = \phi V_n \tag{7}$$

Where:

- ϕ is the strength reduction factor, normally taken as 0.75 for shear.
- V_n is the nominal shear strength of the beam, which is calculated as the sum of the concrete's shear strength (V_c) and the steel shear strength (V_s):

For the beam under consideration, the calculated allowable shear is

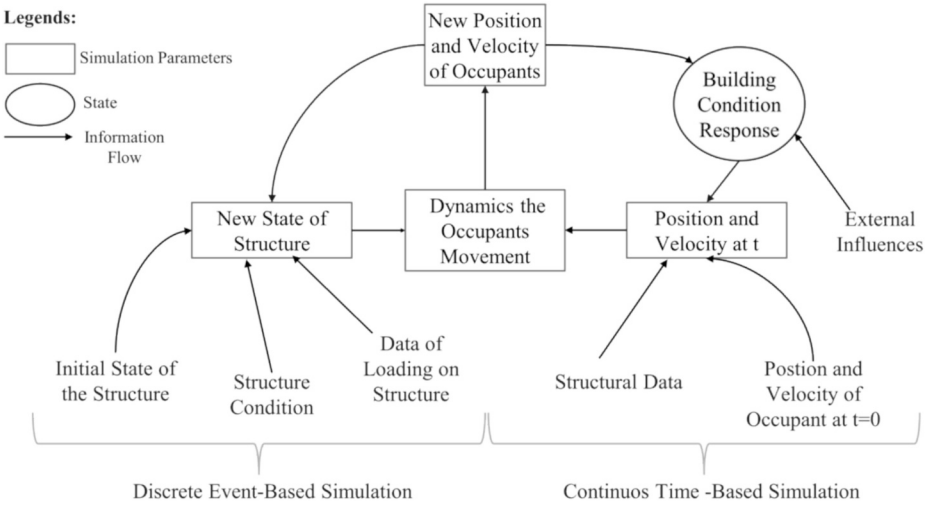
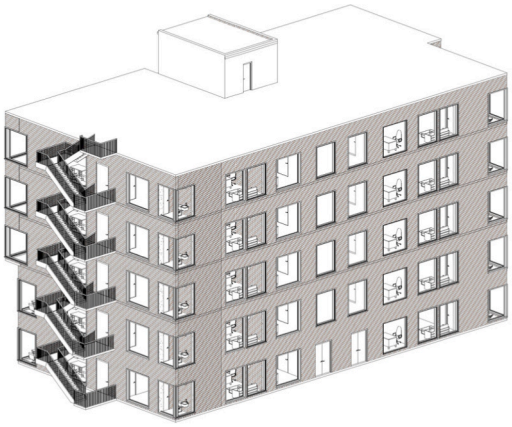
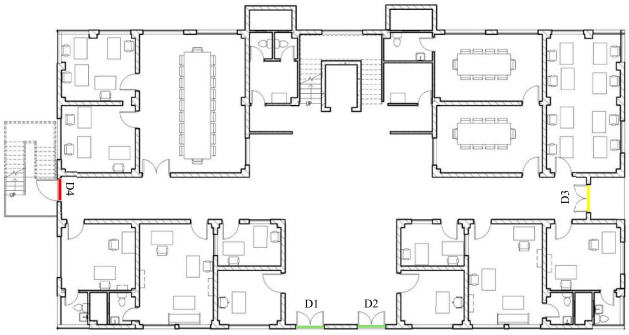


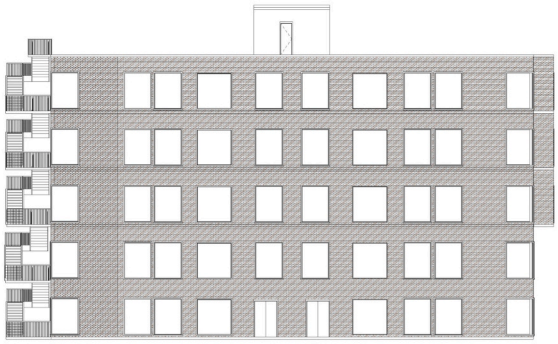
Fig. 8. Pathfinder logic diagram of the simulation process.



a: 3D Model of the building



b: Ground floor plan



c: South view

Fig. 9. Different view of a building under study. (The green color represents the normal passage; the yellow color shows the passage of the back door, while the red color line represents the emergency evacuation passage). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

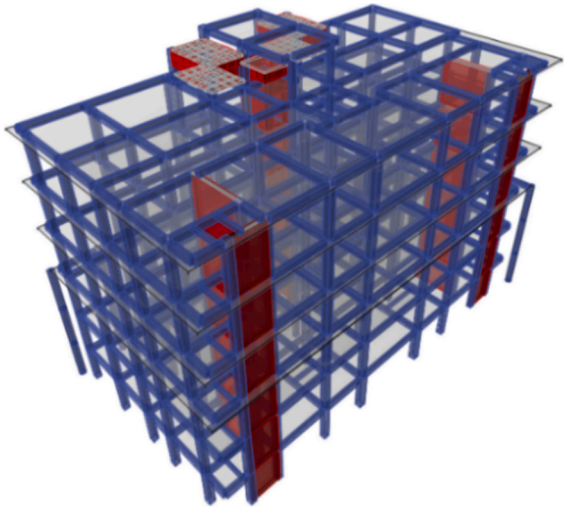


Fig. 10. ETABS structural model (3D).

$V_{\text{allowable}} = 83.18$ KN.

Similarly, the maximum allowable moment ($M_{\text{allowable}}$) is calculated by Eq. (8):

$$M_{\text{allowable}} = \phi M_n \tag{8}$$

Where: $\phi = 0.90$ for flexure

- ϕ is the strength reduction factor, normally taken as 0.90 for shear.
- M_n is the nominal flexural strength of the beam

In this case, $M_{\text{allowable}}$ is calculated to be 308.36 KN-m.

Furthermore, the allowable deflection is determined based on BCP by using the formula given in Eq. (9):

$$\text{Allowable Deflection} = \text{L}/240 \tag{9}$$

Where:

- L is the span length of the beam. For the given beam, the allowable deflection 10.16 mm.

Under the applied load combination, the actual values for the beam are as follows:

The seismic forces during the earthquake event have created conditions where the moment and shear forces acting on the beam above the back door exceed its design structural capacity. This ultimately leads to beam failure as shown in Table 2.

The structural analysis conducted using ETABS software yielded several key results as illustrated in Fig. 12, including base shear and story drift values in both the X and Y directions. The base shear graphs represent the total horizontal forces acting on the structure due to seismic loads in both the X and Y directions. Base Shear FX refers to forces acting in the X-direction, while Base Shear FY refers to forces in the Y-direction. The graphs illustrate how seismic loads in both directions impact the structure, helping to determine the forces transferred to critical structural elements like foundations, columns, and bracings.

The story's drift results show the relative displacement between floors due to seismic forces. The story drift in X direction due to

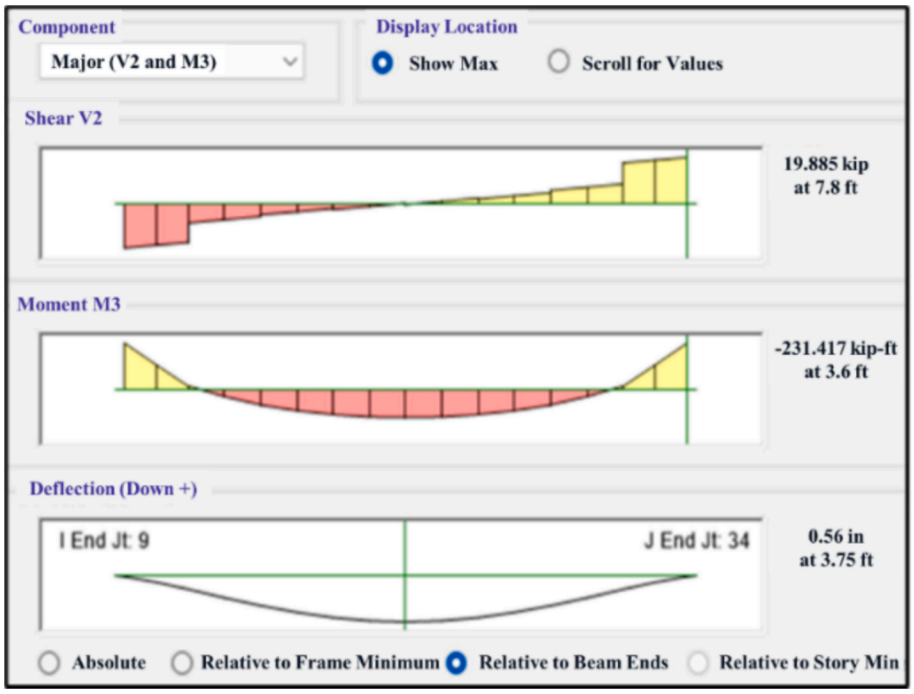


Fig. 11. Shear force and bending moment diagram of beam under loading.

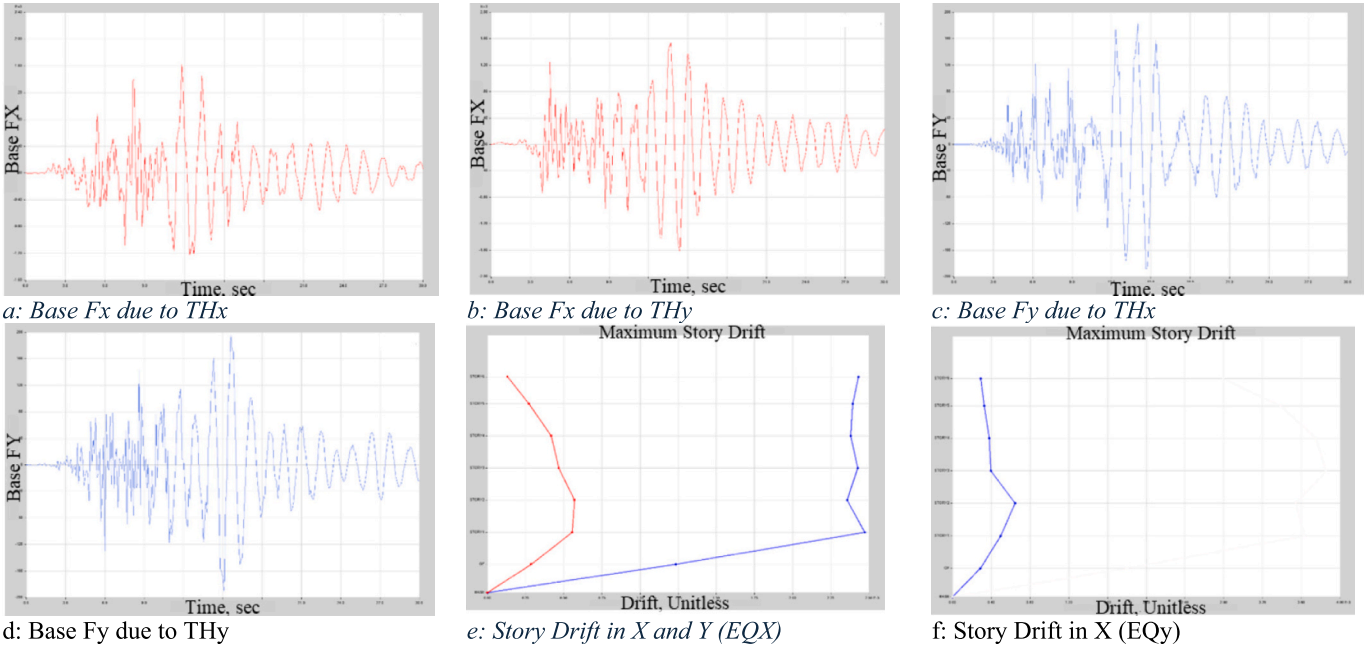
Table 2
Comparison of allowable and actual shear, moment and deflection of beam above back door.

Parameters	Shear (KN)	Moment (KN-m)	Deflection (mm)
Allowable	83.18	308.36	10.16
Actual	88.45	313.68	14.22

earthquake loading in Y direction assesses the drift under Y-direction loads to ensure it remains within safe limits. The story drift Story Drift in X and Y Direction graph compares drifts in both directions due to components of earthquake loading in X direction, highlighting asymmetry and torsional effects that arise from uneven stiffness or mass distribution in the structure.

5.2. Evacuation simulation

Pathfinder [62] is an advanced egress simulation software that utilizes agent-based modelling to simulate occupant movement during evacuation scenarios. It leverages steering behaviors to accurately represent how individuals navigate through space, making it a powerful tool for analyzing and optimizing building evacuation strategies. It has been extensively applied in evacuation studies for high-rise and complex



a: Base Fx due to THx *b: Base Fx due to THy* *c: Base Fy due to THx*

d: Base Fy due to THy *e: Story Drift in X and Y (EQX)* f: Story Drift in X (EQy)

Fig. 12. Graphical representation of results from ETABS simulator.



Fig. 13. Ground floor view in from path finder.



Fig. 14. Contours representing the time required to exit the building.

environments [51], offering advanced agent-based simulations that dynamically adapt to environmental changes such as blocked exits or structural failures. This adaptability is critical for incorporating DE triggers from ETABS. Fig. 13 represents the ground-floor view of the model from Pathfinder, illustrating the positions of occupants. It also shows the trigger created at $t = 14$ s, based on information from the ETABS simulation, which indicates that the beam above the back door is expected to fail at this time.

Fig. 14 illustrates the Pathfinder simulation output for case 2 as detailed in Table B1 of Appendix B: In this scenario, both the main doors and the emergency evacuation door were utilized for evacuation. The output shows the evacuation times for people exiting the building, with color gradients representing time in seconds. The blue areas indicate the quickest evacuation paths, with times as low as 50 s, while the warmer colors (yellow, orange, and red) represent slower exit times, reaching up to 450 s.

The building, designed to accommodate up to 170 occupants, was analyzed using Pathfinder simulations across various evacuation scenarios to evaluate evacuation times. Data obtained from the building's Human Resources office provided detailed information, including the distribution of male, female, elderly occupants, and individuals using wheelchairs, as summarized in Table 3. Additionally, Table 3 outlines the evacuation speeds [63] utilized in the modelling process and specifies the collision response times for occupants, ensuring a comprehensive representation of evacuation dynamics.

Table 3
Evacuation characteristics of building occupants.

	Adults		Elderly	People in Wheelchair	Assisting those on wheelchair
Category	Men	Women			
Age (years)	20 to 50	20 to 50	50 to 65	35 to 60	20 to 50
Numbers	95	41	26	8	8
Average evacuation Speed (m/s)	1.14	1	0.75	0.48	0.48
Collision response time (s)	1.5	1.5	1.12	1.5 to 1.12	1.5
Emergency Awareness	Evacuation drills every 2 months				

In a normal evacuation scenario, all 170 people exited through the main doors in 216.4 s, as mentioned in Appendix B, Table B1. When utilizing emergency exits and the elevator, evacuation time decreased to 187.7 s. Blocking the stairs at 9 m increased the evacuation time to 228.3 s, with all occupants using emergency exits. Incorporating the back door in the process of evacuation with the emergency evacuation door resulted in an evacuation time of 232.1 s, which increased to 234.9 s when the back door was closed 14 s into the evacuation. The emergency-only evacuation took the longest, with a time of 259.6 s, while the combined use of normal exits and the elevator resulted in the longest time of 375.9 s.

On holidays the building has 82 occupants. In a normal evacuation scenario, all 82 people exited through the main doors in 104.8 s, as shown in Appendix B, Table B2. Using emergency exits reduced the evacuation time to 94 s. Blocking the stairs at 9 m increased the evacuation time to 110.6 s, with all occupants relying on emergency exits. Including the back door resulted in an evacuation time of 105.5 s, which increased to 129.1 s when the back door was closed 14 s into the evacuation. An evacuation using only emergency exits took 110.5 s, while combining normal exits and the elevator resulted in the longest evacuation time of 161.2 s.

Fig. 15 illustrates the evacuation dynamics of individuals through different doors under various conditions, offering insights into crowd movement behavior. In Fig. 14a, the smooth cumulative evacuation curve illustrates evacuation dynamics achieved by utilizing normal doors, emergency exits, and elevators, effectively minimizing bottlenecks. While Fig. 14b and c reveal sharper flow rate peaks, indicating surges and congestion caused by uneven occupant distribution. In Fig. 14d, the cumulative number of occupants remaining inside the building decreases steadily until $t = 14$ s, where a slope change reflects redistribution triggered by the back door closure. This event is mirrored in Fig. 14f by a sudden drop in back door flow rates, and in Fig. 14e by increased fluctuations at the EE door, indicative of congestion from rerouted occupants. Similarly, Fig. 14g, h, and i demonstrate similar patterns for a scenario involving 82 occupants within the building.

5.2.1. *Theoretical integration with INTO-CPS framework*

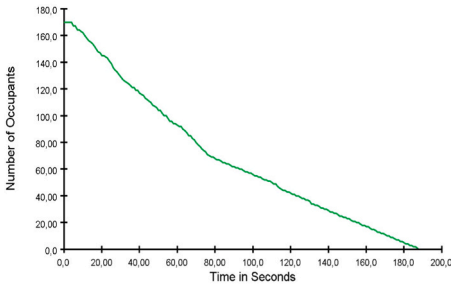
While this study demonstrates manual co-simulation process, future adoption of INTO-CPS will enable automated synchronization, minimizing errors and enhancing simulation reliability. The INTO-CPS framework [64] provides a foundation for automating co-simulation processes, addressing the limitations of manual data exchange between ETABS and Pathfinder. By leveraging the Functional Mock-up Interface (FMI) standard, INTO-CPS facilitates seamless synchronization and data exchange between DE and CT simulations.

In this context, structural analysis data from ETABS (e.g., deformation, failure points) can be encapsulated as a Functional Mock-up Unit (FMU)[1] and integrated with Pathfinder's evacuation dynamics through the INTO-CPS Co-Simulation Orchestration Engine (COE). The COE ensures precise time-step alignment and consistent data flow, enabling real-time updates to evacuation models based on structural events.

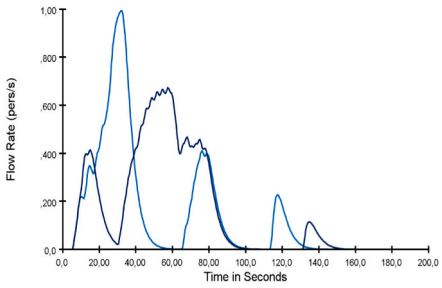
5.3. *Case study-driven development of the BSIC*

The development of the BSIC is deeply rooted in the practical challenges and insights gained from the case study, which examined emergency evacuation in the five-story building. By addressing the

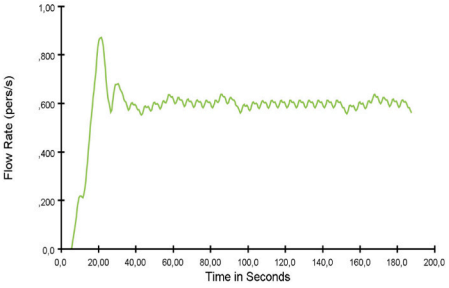
[1] Many projects exist for converting software tools into FMU via a wrapper, such as PythonFMU3: pythonfmu3 · PyPI.



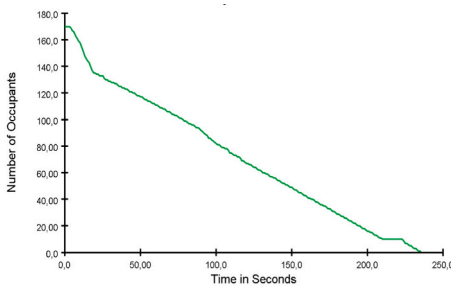
a: Remaining number of people when both normal and E.E doors are used with elevators (with 170 occupants)



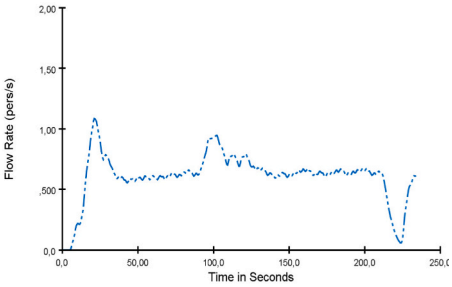
b: Flow rate through main doors when both normal and E.E doors are used with elevators (with 170 occupants)



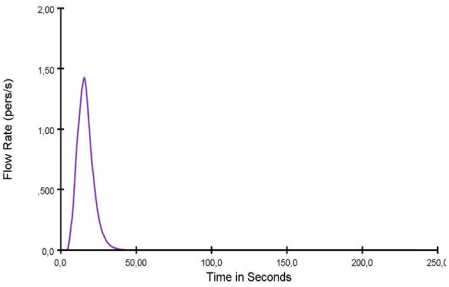
c: Flow rate through E.E door when both normal and E.E doors are used with elevators (with 170 occupants)



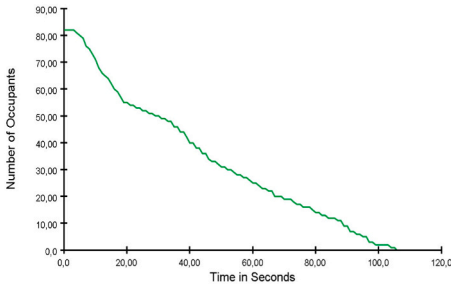
d: Remaining number of people from E.E and back doors and back door is closed at t=14 sec (with 170 occupants)



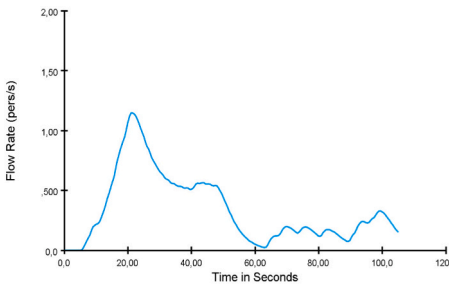
e: Flow rate from E.E door when both E.E and back doors are used, and back door is closed at t=14 sec (with 170 occupants)



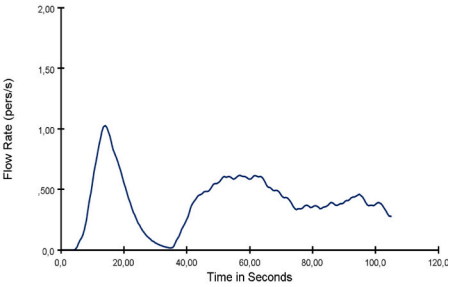
f: Flow rate from back door when both E.E and back doors are used, and back door is closed at t=14 sec (with 170 occupants)



g: Remaining number of people from E.E and back doors (with 82 occupants)



h: Flow rate from E.E door when both E.E and back doors are used (with 82 occupants)



i: Flow rate from back door when both E.E and back doors are used (with 82 occupants)

Fig. 15. Evacuation dynamics including number of occupants, flow rates, and peak flow under various scenarios (E.E = Emergency Evacuation).

integration of multiple simulation models—structural integrity and evacuation simulations—the case study offers a critical testing ground for development and refining the BSIC framework. The structural analysis conducted using ETABS software, combined with evacuation simulations performed via Pathfinder, highlighted the challenges in co-simulation frameworks within the building industry, as outlined below.

Five critical challenges were identified:

- ETABS models DEs, such as beam failure at $t = 14$ s, whereas Pathfinder requires continuous updates to accurately represent occupant movement.
- The distinct data formats and solvers utilized by ETABS and Pathfinder introduced significant challenges in automation, often requiring manual data transfers that increased the risk of error.
- Misalignments in the synchronization of time steps resulted in discrepancies between the timing of structural failures and occupant responses.
- Variability in underlying assumptions and computational paradigms further compounded integration issues. Precise alignment of structural deformation rates with human flow dynamics necessitated adjustments.
- Identifying the root causes of inconsistencies within the integrated simulation was complex, as misalignments could arise from assumptions embedded in either the structural or behavioral models.

These findings underscored the urgent need for a standardized co-simulation framework that could accurately manage the interoperability between different simulation paradigms. The failure of a beam at a specific moment (a DE event) demonstrated the requirement for immediate updates to the occupant movement paths (a CT process). This scenario showcased how critical time-step misalignment, data synchronization issues, and model interoperability could compromise the simulation's reliability. By identifying and documenting these issues, the case study provided foundational insights for the design of BSIC, aimed at addressing these challenges systematically.

BSIC is thus developed to formalize the method for documenting and sharing essential simulation parameters, including state vectors for CT processes and event triggers for DE events. This structured approach ensures that critical information flows consistently between different models, facilitating smoother co-simulation. The result is a robust framework that bridges communication and data synchronization gaps, ensuring that co-simulated models can interact with minimum errors. With the foundational challenges highlighted through the case study, the next step is to explore the detailed structure and the key attributes of BSIC that define its functionality and application in a complex co-simulation environment in the building industry. The comprehensive BSIC for ETABS simulations is detailed in Table S1 of the supplementary material.

5.4. Overview of the BSIC framework and its key attributes

BSIC provides a comprehensive framework for documenting and managing the various attributes of co-simulation models in the building industry, offering a standardized method for defining and analyzing simulation parameters. When applied to complex building co-simulations, such as those discussed in Section 1.1, the BSIC attributes play a critical role in addressing many of the integration, synchronization, and data exchange issues that typically arise when combining multiple simulation models in the building industry. These attributes facilitate a clearer understanding of the structural, dynamic, and human aspects of a building during emergency situations such as evacuations or structural failures.

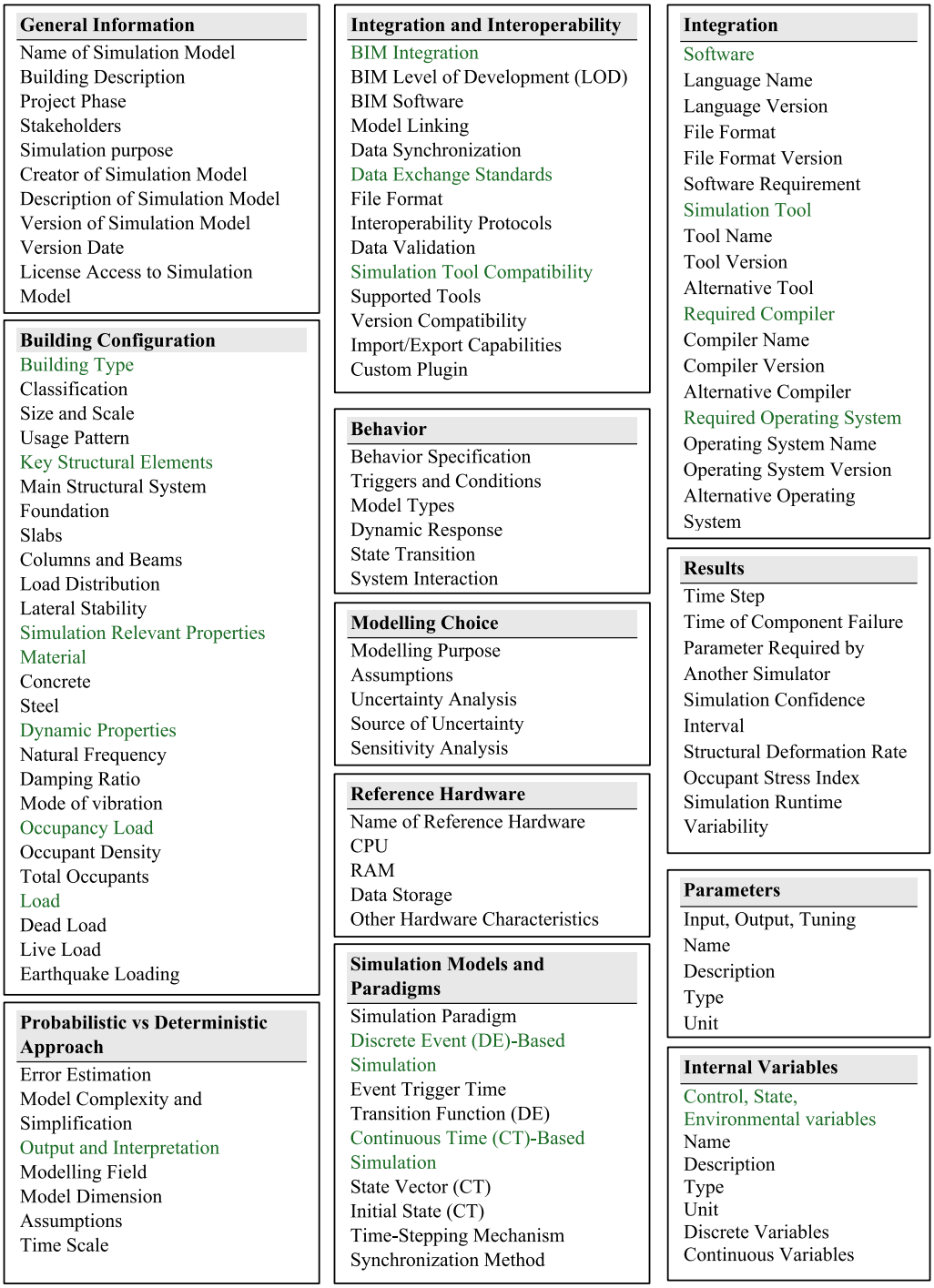


Fig. 16. Attributes of BSIC.

Fig. 16 outlines the classification of attributes within the BSIC framework, which is used to evaluate and integrate structural evacuation analysis simulation models. It categorizes attributes into several key areas: general information, integration and interoperability, building configuration, behavior, results, internal variables, and reference hardware. Each category addresses essential factors such as software requirements, data exchange standards, building elements, dynamic properties, control parameters, and hardware compatibility. This classification serves as a comprehensive guide for structuring simulation data and ensuring interoperability in building performance assessments.

Although simulation models encompass various categories of attributes, as shown in Fig. 16. However, this section is specifically focused on building-related categories to facilitate a targeted examination of structural and evacuation co-simulation.

– ***General information and simulation models***

The general information section of the BSIC provides foundational details, including the name and purpose of the simulation model, the building type, and the project phase. By clearly defining these elements, all stakeholders, e.g., architects and engineers, agree regarding the simulation's objectives. For example, an emergency evacuation simulation may have a drastically different focus and set of parameters compared to a model that simulates long-term energy efficiency. The stakeholders' attribute ensures that the relevant parties are informed about the scope and purpose, fostering collaboration and reducing ambiguity.

The purpose of the simulation is also crucial in guiding how different simulation units are developed and synchronized. For example, to simulate emergency evacuations, the simulation model will focus on dynamic movement patterns, incorporating human behavior and social interactions. Defining this purpose early helps streamline the integration of various simulation paradigms.

– ***Building configuration and structural elements***

The building configuration attributes on the BSIC, such as building type, size and scale, and key structural elements, provide essential inputs for accurate simulations. By defining these attributes in a structured format, the BSIC card allows different simulation models (e.g., for structural analysis, evacuation dynamics) to work with a consistent set of building parameters.

For example, in a simulation focused on an office building's emergency evacuation, the main structural system and lateral stability are crucial components. These elements ensure that the building can withstand dynamic loads (e.g., from seismic activity) while facilitating occupant movement toward evacuation points. The reinforced concrete frame structure with shear walls for lateral stability is a typical configuration that must be represented consistently across different models in a simulation. The integrity of the structural elements, particularly during emergencies, affects evacuation success rates, making them a vital factor in simulation accuracy. The BSIC standardizes how this information is captured and shared, promoting consistency across models.

The building usage pattern attribute further helps define how the simulation should account for varying occupancy rates and occupant behavior. As the building is used primarily during office hours with occasional weekend activity, it has different evacuation patterns compared to a high-occupancy building like a school or hospital. Incorporating these patterns into the simulation helps refine both the dynamic load assumptions and the human factors, ensuring that the simulation aligns with real-world conditions.

– ***Dynamic properties and occupancy load***

Attributes such as natural frequency, damping ratio, and mode of vibration are critical for simulations involving structural response to dynamic loads, such as earthquakes or wind forces. These attributes feed directly into DE and CT simulation models, allowing for an accurate representation of how the building behaves under various stress conditions. For instance, an initial state CT simulation may focus on the structural integrity of the building at $t = 0$, while subsequent states reflect progressive structural deformation.

In evacuation-focused simulations, the occupancy density and total occupants' attributes are essential in calculating evacuation times and identifying potential bottlenecks. These attributes provide input for DE-based simulations, where events like the opening of exits or the movement of occupants are triggered based on discrete conditions. These elements of simulation align with the need to incorporate human factors, a key issue in the building industry. Unlike mechanical systems in automotive or aerospace industries, where human interaction is minimal, buildings must account for the unpredictable nature of human movement and decision-making during emergencies.

– ***Integration with BIM and data exchange***

One significant challenge is the lack of standardized communication between simulation models, particularly when models are developed by different experts using different tools. The BIM integration attributes on the BSIC help to address this by promoting interoperability between BIM and various simulation platforms. The BIM Level of Development (LOD) indicates how detailed the BIM model is, allowing the simulation team to gauge the accuracy of the input data.

By facilitating direct interaction between BIM models (e.g., using Autodesk Revit) and simulation tools (e.g., Pathfinder, Pyrosim), the BSIC ensures that the physical space represented in the simulation accurately reflects the real-world building. The model linking and data synchronization attributes are important for maintaining consistency across simulations using different tools or platforms. For instance, exporting the BIM model as a DWG or IFC file ensures that critical structural and architectural elements (e.g., exits, stairways, walls) are correctly positioned and modelled for evacuation simulations. This synchronization helps mitigate the problem of data exchange inaccuracies.

– ***Simulation models and paradigms***

BSIC allows documentation of whether a DE or CT approach (or a hybrid model) is being used, and how these models are synchronized. The use of a time-stepping mechanism like the Jacobi algorithm ensures that both DE and CT simulations remain aligned during the co-simulation process, reducing the risk of time-step misalignment and event-trigger errors.

Furthermore, attributes like state vector (CT) and event trigger time (DE) allow for precise tracking of both continuous structural changes (e.g., deformation under load) and DEs (e.g., structural failure when loads exceed capacity). This structured approach to simulation documentation provides a clear framework for resolving synchronization challenges, thus improving accuracy and reliability in hybrid co-simulations.

The attributes defined in the BSIC serve as essential building blocks for addressing the complexities of co-simulation in the building industry. By providing a standardized method for capturing building configurations, structural elements, occupancy patterns, and dynamic properties, BSIC promotes interoperability, data consistency, and accurate synchronization between DE and CT simulations.

Table 4 presents some of the key attributes from the **BSIC** that are important for the seamless integration of DE and CT simulation models. BSIC acts as a standardized framework that comprehensively describes the key characteristics of each simulation model, enabling effective interoperability across cross-domain collaboration. By systematically documenting attributes such as model type, event trigger mechanisms, state variables, time synchronization methods, and data formats, the BSIC facilitates interoperability between diverse simulation models. In

Table 4
Key BSIC attributes that facilitate the integration of discrete-event (DE) and continuous-time (CT) simulation models.

BSIC Attribute	Description	Purpose for DE and CT Integration	Details for Integration
Model Type (DE or CT)	Specifies whether the simulation model follows a DE or CT approach.	Identifies whether the model represents DEs (e.g., structural failure, door closure) or continuous changes (e.g., deformation, occupant movement) to enable seamless interaction during co-simulation.	Structural simulation includes CT for continuous deformation tracking and DE for structural failures. Occupant simulation uses CT for movement and DE for events like passage blockage or door closure.
Event Trigger and Timing	Defines specific events in DE models (e.g., structural component failure, door closure) and the conditions under which they occur, including time-based triggers.	Helps synchronize key events in the DE model with continuous processes in the CT model, ensuring that real-time responses such as occupant evacuation or structural adjustments are handled appropriately.	For structural models, the failure of the beam is triggered at t = 14 s. For occupant models, passage blockage occurs after a structural failure, rerouting occupants through alternate paths.
State Variables (CT)	Describes the continuously changing variables in CT models, such as deformation rates or occupant positions, which evolve over time and need to be updated in real-time.	Ensure continuous feedback from CT models is integrated into DE models so that events (e.g., failure, door closure) respond to real-time conditions, maintaining consistency in simulation outcomes.	Continuous deformation in the structure is fed into the DE model to trigger structural failures when stress thresholds are crossed, which in turn affect evacuation paths in real-time occupant movement.
Time Synchronization Method	Details the method used to synchronize DE and CT models, ensuring that both operate on compatible time scales and prevent mismatches during co-simulation.	Critical for aligning DEs (e.g., failure at t = 14 s) with ongoing continuous changes (e.g., deformation, occupant movement), ensuring a smooth transition between discrete and continuous simulation phases.	The Jacobi method is used to synchronize time-steps between structural failures (DE) and continuous occupant movement (CT), ensuring that events like passage closure align with the structural and evacuation dynamics in real-time.
Input/Output Data Formats	Defines the data formats used for exchanging information between DE and CT models, ensuring compatibility in variables such as structural load, occupant position, and event triggers.	Facilitates smooth data exchange across different simulation models, preventing data conversion errors that could disrupt the accuracy of co-simulation results, especially when events are tied to real-time structural changes.	BIM-derived data (in IFC format) is used for model interoperability. Structural loads and occupant position data are exchanged between DE and CT models using standardized formats to maintain consistency across simulations.
Interoperability Protocol	Specifies the communication protocols or standards used for co-simulation, ensuring that DE and CT models are interoperable and can share real-time data without errors.	Ensures that both DE and CT models can communicate seamlessly during co-simulation, allowing real-time updates and interactions between structural failures (DE) and occupant responses (CT).	Functional Mock-up Interface (FMI) can be used as the standard for integrating structural and evacuation models, ensuring that structural deformation and failures (DE) are communicated to the occupant movement model (CT) in real-time.

the integration of DE and CT simulations—where structural failures (DE) must be aligned with continuous occupant movement (CT), or structural deformations (CT) with discrete evacuation events (DE)—the BSIC ensures that key events are synchronized in real-time. This standardization is crucial for maintaining coherence between models developed independently by different stakeholders, allowing for consistent data exchange and reducing integration errors that could arise from mismatched time-steps or incompatible data formats.

6. Discussion

The RQ focused on how effectively a framework for building simulation units improves communication between simulation models in a co-simulation process. The case study demonstrated that the BSIC framework successfully addresses this by facilitating the standardization of data exchange formats, synchronizing time steps, and ensuring accurate event propagation between DE and CT models. Specifically, the integration of structural failure events (DE) with continuous occupant movement (CT) in the evacuation scenario showcased BSIC's ability to minimize integration errors and enhance simulation reliability.

In this section, we critically examine the theoretical and methodological foundations that have informed the development of the BSIC framework. We begin by reviewing existing co-simulation standards and methodologies, assessing their successes and limitations in enabling the development of BSIC. By identifying key gaps in communication and interoperability, we illustrate how these insights directly influenced the design of BSIC to address the distinct and complex requirements of the building sector. Subsequently, the discussion highlights the framework's contribution to synchronizing DE and CT simulations, thereby enhancing cross-domain collaboration and ensuring data consistency.

6.1. Strengthening concepts through existing co-simulation standards

Our initial contribution to the RQ is to understand the existing knowledge on co-simulation standards and techniques across various fields, which have contributed to the development of the BSIC framework. For instance, the standards such as FMI [22], helped in the development of BSIC by providing insights into advanced model exchange and co-simulation methodologies. FMI's structured methods for standardized model data handling, event synchronization, and layered standards enabled BSIC to effectively include the information that helps to coordinate diverse simulation models and promotes compatibility across tools and data formats.

The HLA standard [34] with a focus on promoting interoperability in complex, distributed simulation environments, has influenced the BSIC approach for managing diverse building simulations, as it enables standardized model communication and data exchange across various simulation models. This was particularly helpful in defining BSIC structure to enable it to serve as a bridge across domains like structural, environmental, and evacuation modelling, allowing for seamless integration and interaction among models developed by different stakeholders. HLA process of synchronizing federates in distributed simulations motivated BSIC to adopt similar methods for coordinating structural and evacuation simulators. The concept of a runtime infrastructure in HLA inspired BSIC's approach to managing interactions between hybrid simulation models.

Similarly, the SSP standard [25], with its structured approach to model parameterization and exchange, informed the development of BSIC detailed specification of simulation inputs, outputs, and properties. By focusing on tool-agnostic system packaging and parameter management, SSP provided a foundation for BSIC to encapsulate diverse model attributes, such as building dimensions, structural elements, and load specifications, thereby facilitating data consistency and reusability across different simulation platforms.

Moreover, the MIC [27], initially used in the automotive industry, served as a foundation for BSIC's structured format. While MIC provides a structured way to document simulation attributes, it does not fully address the complexities of human dynamics and long-term structural behavior inherent in the building industry. Our literature review identified the necessity of incorporating building-specific features, such as occupancy load variability and adaptive structural analysis, into BSIC.

6.2. Justification of BSIC in co-simulation framework in the building industry

The BSIC is specifically designed to tackle the complex challenges of integrating multiple unified simulation models within the building industry, a sector characterized by long-term performance requirements and diverse functional needs. These simulations require the integration of multiple tools, each focusing on different aspects of building performance, including structural integrity, energy consumption, and occupant behaviors. This aligns with the second contribution of the paper to address the RQ, which facilitates interoperability across simulation units by standardizing parameters, variables, and behavioral expectations for seamless integration. Furthermore, it directly supports the third contribution by enabling solutions to the challenges of time-step alignment between simulation paradigms, ensuring accurate and reliable co-simulation outcomes in the building industry.

6.2.1. Improved communication and data exchange

A key topic in the development of BSIC is enhancing communication between different simulation models by standardizing model attributes. This reduces the likelihood of errors during data exchange and helps to ensure that different models can communicate seamlessly. The idea of a BSIC, inspired by the MIC from industries like automotive and aerospace, was adapted to the building context to account for the building-specific requirements of human dynamics, long-term usage, and diverse building types.

Yeung et al. [38] and Cellura et al. [39] identified that while frameworks exist to facilitate model exchange, there are significant challenges in maintaining interoperability, especially when models use different geometric representations or data formats. The BSIC will be helpful in this by providing a uniform method for capturing and documenting simulation attributes, such as data exchange formats and model parameters, which improves the reliability of cross-domain co-simulations (as depicted in Fig. 16).

In the case study the application of BSIC demonstrated its effectiveness in addressing the challenges of communication and data exchange between ETABs and Pathfinder simulation models. It ensured that evacuation paths adapted to structural changes, thereby reducing integration errors. The outcome of the case study supports RQ by showcasing BSIC's role in enhancing co-simulation reliability in the building industry.

6.2.2. Synchronization of continuous-time and discrete-event simulations

The integration of DE and CT simulations is one of the most significant contributions of BSIC to address the RQ, ensuring accurate event propagation and enhanced co-simulation reliability. In co-simulation environments, these two paradigms are essential for capturing both the continuous dynamic changes in a building and the DEs that occur at specific points in time, such as structural failures or evacuation events.

Yang et al. [35] provide an excellent example of a HLA for co-simulation, focusing on infrastructure resilience. Their study highlights the challenges of synchronizing time steps and managing communication between systems with different event frequencies. Additionally, they point out the difficulties faced by stakeholders from various organizations in adopting such a framework. The BSIC helps to address these synchronization challenges by introducing key attributes like state vector (CT) and event trigger time (DE). These attributes allow precise tracking of continuous structural changes and DEs within a co-simulation environment. By standardizing the method for synchronization, the BSIC facilitates the alignment of DEs—such as beam failures—are aligned with the continuous deformation of the structure, facilitating accurate real-time interaction between DE and CT models.

This synchronization between DE and CT models is especially critical in emergency evacuation scenarios as described in the case study. For example, when a structural component fails due to earthquake loads (a DE event), the evacuation simulation must dynamically adjust to account for this structural change (a CT process). This interplay between DE and CT processes is illustrated in Fig. 11, which depicts structural failure, and Figs. 15d, 14e,f, which demonstrates how evacuation paths adapt dynamically in response to these failures. The BSIC guarantees that these events are managed smoothly and accurately, minimizing the risks of time-step misalignment and event-trigger errors.

6.2.3. Handling occupancy and human dynamics in co-simulation

Unlike industries like automotive or aerospace, the building sector requires simulations to account for human behavior and social dynamics, especially in emergency situations. The case study demonstrates that BSIC ensures that human factors, such as occupant movement, crowd behavior, and evacuation patterns, are integrated into the co-simulation framework in a structured way addressing the RQ by improving interoperability in co-simulation processes. This is essential for modelling realistic evacuation scenarios where human responses to emergencies vary significantly.

The inclusion of attributes like occupancy load and behavioral Patterns in BSIC (as highlighted in Fig. 16) facilitates the synchronization of both structural simulations and human dynamics. This integration allows for more accurate predictions of evacuation times and better emergency planning, aligning with the goals of prior works like those of Benzaama et al. [40], who used co-simulation techniques to model energy efficiency and occupant comfort in buildings.

6.2.4. Scalability and applicability of BSIC

The scalability and applicability of BSIC framework were demonstrated through the case study of a five-story office building in Islamabad, Pakistan.

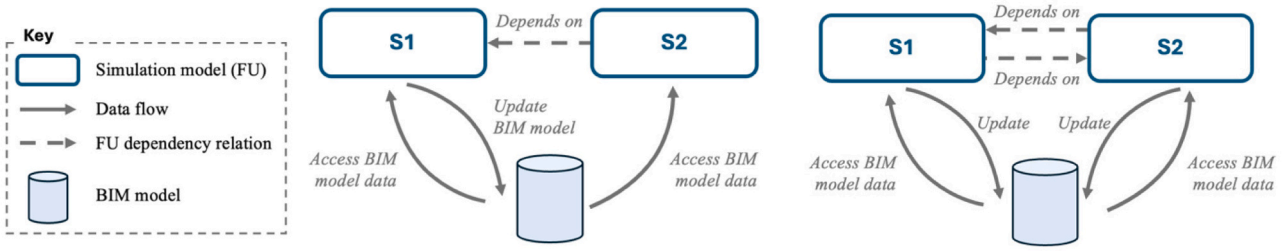
6.2.4.1. Scalability. The BSIC framework is designed to address integration challenges in co-simulation for the building industry. With respect to the *RQ*, scalability refers to whether the means of integrating simulation models via the BSIC framework enables co-simulations to execute within a practical period of computational runtime as the size and complexity of the building model (i.e. the *input* to the co-simulation) increases to the scale of real-world building models.

The case study demonstrated that BSIC, when integrating two simulators, can execute co-simulations that indeed scale to handle buildings of varying sizes and complexity, with the combination of ETABS and PathFinder taking between 97,4 s and 118,3 s depending on the scenario (Appendix B). The five-story office building presented a moderately complex scenario involving multiple evacuation routes, structural elements, and dynamic loads. This indicates that co-simulation is viable at real-world scales, thus addressing the RQ with respect to facilitating the integration of diverse simulation models in practice.

In principle, the computational runtime needed to execute a co-simulation is at least as long as the sum of the computational runtimes needed to execute each simulation model independently. Executing a co-simulation (e.g. using the Jacobi method) incurs additional runtime due to:

- Pausing and resuming each simulation model multiple times;
- Time needed by simulation models to reconfigure themselves when the building state changes while the simulation is paused (e.g. structural failure causes a change to the paths, which may require the occupant simulator to re-generate internal data structures used to compute occupant paths);
- Requesting event information from each simulation model, i.e. requesting the time when the "next important event" occurs from each simulation model.

The dependencies between simulation models are anticipated to play a key role in the computational runtime required to execute a co-simulation (in addition to accuracy). In the case study, the ETABS



(i) Acyclic dependency between simulation models, due to only one simulation model (S1) modifying the BIM model in a way that impacts the other simulation model (S2).

(ii) Cyclic dependencies between simulation models due to both S1 and S2 modifying the BIM model in ways that impact the other simulation model.

Fig. 17. Acyclic and cyclic dependencies between simulation models.

simulator impacted the BIM model by calculating structural failures, while the PathFinder occupant simulator did not impact the BIM model with respect to stress forces that would otherwise have been fed back into ETABS to re-calculate new potential structural failures, as illustrated in Fig. 17 (i). This meant that the ETABS simulator could be executed to derive time points at which milestone (discrete) events occurred, which were then delivered to PathFinder; no further information about the movement of the occupants needed to be (indirectly) fed back into ETABS. In contrast, if the number and collective weight of occupants moving through the building was sufficiently high, and the structure sufficiently compromised, then feedback scenarios could be co-simulated, resulting in cyclic dependencies between ETABS and PathFinder, as illustrated in Fig. 17 (ii). This would result in more simulator "pauses", simulator data structure updates, and so on. Investigating the runtime impact of having more complicated cyclic dependencies between simulators in relation to the size of the building model and other scenario parameters (such as number of occupants) is an important topic for future research.

6.2.4.2. Applicability. The BSIC framework is applicable across various scenarios in the sector. For example, it has been demonstrated to support emergency response planning by enabling co-simulation of structural integrity and evacuation dynamics, thereby ensuring comprehensive safety analyses. Its compatibility with widely used BIM platforms, such as Autodesk Revit, enhances its utility by enabling integration with existing architectural and structural models. In the case study, the office building's evacuation dynamics were analyzed under various conditions, including full occupancy of 170 occupants and partial occupancy of 82 occupants. The results demonstrated that BSIC could effectively adapt to different occupancy patterns, evacuation routes, and structural configurations. This addresses the RQ with respect to encouraging and promoting co-simulation owing to its *applicability* in exploring scenarios more comprehensively than is possible with simulation models used in isolation.

Despite its potential, the framework has certain limitations. Its reliance on standardized data formats, such as IFC and XML, restricts its use with simulation tools that do not support these formats. Furthermore, BSIC's dependence on manual data integration, as evidenced in case studies, limits its effectiveness in scenarios requiring rapid deployment or high automation levels.

7. Conclusions

This paper addressed the research question of enhancing communication between simulation models in co-simulation processes for the building industry by developing the Building Simulation Identity Card (BSIC) framework through a detailed case study. This case study approach demonstrates how BSIC facilitates the integration of DE and CT simulations, enabling precise and synchronized data exchange between structural analysis and evacuation simulations. Results from the case study show that the BSIC framework supports system-level real-time communication, effectively reducing integration errors and promoting accuracy in co-simulated building performance analysis.

The theoretical contributions of this paper are twofold. First, it presents a comprehensive review of existing co-simulation methodologies across various domains, highlighting critical gaps in communication and interoperability. Subsequently, by considering these gaps, the BSIC framework was developed, marking a significant advancement in theoretical practices. This framework introduces new strategies for the integration and synchronization of models across different domains. By standardizing parameters such as time-stepping mechanisms, event triggers, and behavioral expectations, the BSIC provides a theoretical foundation for cross-domain co-simulations, laying the groundwork for future studies in hybrid simulations across multidisciplinary fields.

In practical terms, the application of the BSIC in a case study focused on an earthquake-induced evacuation scenario demonstrates its effectiveness in aligning DE and CT simulations. This practical application underscores the framework's utility in enhancing emergency preparedness and building safety by enabling accurate data sharing and event alignment between structural failures and occupant movements. Furthermore, the implementation of BSIC significantly advances cross-domain collaboration by providing a unified vocabulary and standardized integration procedures, which are critical in fields such as structural engineering, emergency planning, and digital twin development.

By supporting the creation of DTs with structured and standardized attributes, the BSIC enhances traceability and reliability in data exchanges, contributing to more effective building simulations. The framework addresses the specific needs of the construction sector, enabling the modelling of human dynamics alongside structural behavior, and builds upon standards like the MIC with specialized adaptations for the building industry.

One limitation of this work is the complexity involved in implementing the BSIC framework across diverse tools and simulation environments, which may require significant customization and adaptation efforts from stakeholders. Additionally, while the framework standardizes key attributes, further testing across a broader range of case studies is needed to ensure its scalability and effectiveness in different types of buildings.

Future research will extend dynamic data modelling within a BIM-integrated DT framework to buildings on a large scale, focusing on feedback for non-structural components. This involves developing methods for the seamless integration of simulation data to ensure that each cycle of analysis accurately informs and updates the DT. The future work will focus on establishing a link between the BSIC framework and dynamic data modelling with BIM standards, such as Industry Foundation Classes (IFC), Information Delivery Specifications (IDS), and Building SMART Data Dictionary (BSDD) [62].

Funding

Funding for this Project is Provided by Higher Education

Commission of Pakistan

CRediT authorship contribution statement

Muhammad Farhan Jahangir: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Aliakbar Kamari:** Writing – review & editing, Supervision, Conceptualization. **Carl Peter Leslie Schultz:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Jacobi Algorithm

In this appendix section we present computational details on the Jacobi algorithm, a popular Master Algorithm used by the Co-Simulation Orchestration Engine to execute a co-simulation. The Jacobi Algorithm is particularly useful in scenarios where the system is represented as a collection of loosely coupled subsystems, each contributing to the overall dynamics. The algorithm ensures consistency and stability by advancing all subsystems simultaneously based on the most recent available information, promoting parallelism and modularity. This algorithm is implemented in digital tool chains for facilitating co-simulation such as INTO-CPS.

In the context of the study presented in this paper, the Jacobi algorithm was employed to synchronize discrete-event (DE) and continuous-time (CT) simulations. DE simulations model event-based occurrences such as structural failures, while CT simulations represent ongoing dynamic changes like structural deformation or occupant movement. Effective synchronization between these paradigms is critical to ensure accuracy and consistency in co-simulated outcomes. The Jacobi algorithm's parallel nature and ability to handle independent subsystem computations make it well-suited for co-simulation. By synchronizing time steps uniformly, it ensures the integrity of simulation outcomes, particularly when integrating heterogeneous models like structural integrity and evacuation dynamics.

Figure A1presents a class diagram showing the relationship between FMUs that adhere to the FMI v3.0 standard,[2] and the cosim orchestration engine. Two state computation functions are used in the Jacobi algorithm that are defined in the FMI v3.0 standard:

- *fmi3UpdateDiscreteStates*: used to signal to the FMU that a converged solution of the constraints that define the state of the model is requested at the current time instant. Importantly, one of the returned values is *nextEventTime* which is the absolute time of the next time event t_{i+1} (if it is defined).
- *fmi3DoStep*: request the computation of the FMU's next time step.

When accessing or modifying variable data from FMUs, one of a set of *fmi3Get{VariableType}* (for accessing data) and one of a set of *fmi3Set{VariableType}* (for modifying data) is used.[3] Supported data types include: 32-bit float, 64-bit float, 64-bit unsigned integer, string, Boolean, and so on.

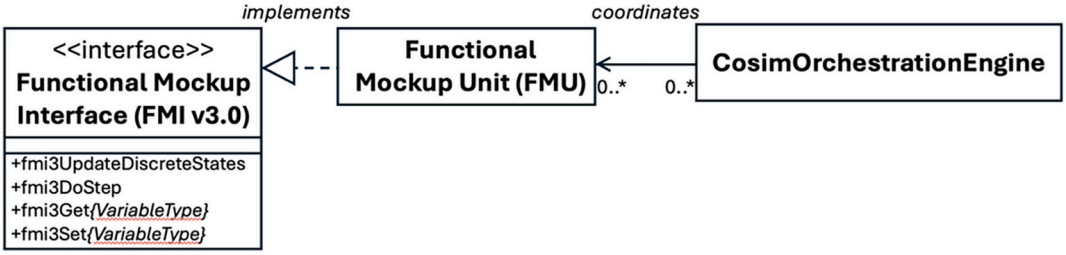


Fig. A1. Class diagram showing how the orchestration engine utilizes the FMI v3.0 standard to implement Master Algorithms, including the Jacobi algorithm.

Class diagram showing how the orchestration engine utilizes the FMI v3.0 standard to implement Master Algorithms, including the Jacobi algorithm.

The Jacobi Algorithm proceeds as follows:

1. **Initialization**: All simulation units (FMUs) are initialized with starting conditions. Each unit operates independently at this stage.
2. **Data Exchange**: At each iteration, each simulation unit computes its state for a fixed time step based on inputs from other units.
3. **Time Synchronization**: The algorithm advances the simulation time uniformly for all units to the smallest common time step.
4. **Convergence Check**: Outputs from all units are compared, and iterations continue until changes between successive iterations are within an acceptable tolerance, ensuring convergence.

A multi-model contains information about which FMU instances exist in a given co-simulation, the initial values of all FMU variables, and the output-to-input port connections between FMU instances. A simulation termination condition is also given, usually as the maximum runtime of the simulation (i.e. the co-sim should terminate once the simulation time has advanced beyond the given termination threshold).

The following pseudo code illustrates an implementation of the Jacobi Algorithm, given a multi-model as input. In our pseudo code algorithm the multi-model is represented as follows:

[2] The FMI v3.0 standard is available online: https://fmi-standard.org/docs/3.0.2/.

[3] This notation convention is prescribed in the standard as a shorthand for a family of functions. We refer the reader to the standard for further details: https://fmi-standard.org/docs/3.0.2/#get-and-set-variable-values.

- Each initial value v of variable x of FMU f is represented as a record [fmu = f, variable = x, initialValue = v]
- Each dependency between an output port *out* of FMU f_1 and input port in of FMU f_2 is represented as a record [outputFmu = f_1, outputPort = *out*, inputFmu = f_2, inputPort = *in*]

We use dot notation for function execution and variable access, and ":="represents variable assignment.

Algorithm JacobiAlgorithm
Input:F (set of FMU instances)
I (set of initial values for each FMU variable)
D (set of dependencies between FMU output and input ports)
Terminate (terminating condition predicate)
Output: None

```
1.  for each i in I:
2.    i.fmu.fmi3Set(i.variable, i.initialValue )
3.  while( ! Terminate() ):
4.    for each d in D:
5.      val := d.outputFmu.fmi3Get(d.outputPort)
6.      d.inputFmu.fmi3Set( d.inputPort, val )
7.    minTNext := inf
8.    for each f in F:
9.      tNext := f.fmi3UpdateDiscreteStates()
10.     minTNext := min( tNext, minTNext)
11.   for each f in F:
12.     f.fmi3DoStep( minTNext )
```

Lines 1–2 correspond to the initialisation phase. Lines 3–12 correspond to the main co-simulation loop that is repeatedly executed until the termination condition is satisfied. Lines 4–6 correspond to the data exchange between FMU instances. Lines 7–10 corresponds to convergence where the time of the next event across all FMUs is determined, *minTNext*. Lines 11–12 correspond to actual simulation where time is advanced across all FMUs *minTNext*.

Appendix B. Appendix

This appendix section provides a comprehensive representation of the results obtained from various evacuation scenarios for the building under study. These scenarios are assessed under two distinct occupancy conditions: full occupancy, involving 170 occupants, and partial occupancy, with 82 occupants. The detailed results encapsulate critical metrics such as evacuation times, flow rates through designated exits, and the distribution of occupant movement across different evacuation paths

The computational runtime taken by ETABS to complete its simulation was 94 s. PathFinder had a computational runtime between 3,4 s and 24,3 s depending on the scenario. Experiments were run on a HP with 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz, 16 GB RAM, and 64-bit operating system.

Table B1
Different evacuation scenarios with 170 people.

		Total number of people	No. of people through main doors	No. Of people through emergency door doors	No. Of people through elevator	No. Of people through back doors	Computational runtime of PathFinder simulator (s)	Simulated total evacuation time (s)	Simulator time of 1st person evacuated (s)	Simulated time of last person evacuated (s)	Min travel distance (m)	Max travel distance (m)
1	Normal evacuation	170	170	0		0	9,9	216,5	4,1	216,4	4	
2	Normal plus emergency with elevator	170	58	112	1	0	11	187,8	4,1	187,7	4	100,5
3	Stairs at 9 m and above is blocked	170	0	170		0	10,2	228,5	4,7	228,3	4,7	105,5
4	Emergency door and back door	170	0	144		26	10,1	232,3	3,5	232,1	3,2	104,4
5	Back door closed at t = 14 s	170	0	153		17	11,6	235	3,5	234,9	3,2	135,5
6	Emergency Evacuation	170	0	170		0	24,3	259,8	4,7	259,6	4,7	101
7	Normal + elevator	170	170	0	54	0	19,4	376	4,1	375,9	4	92,4

Table B2
Different evacuation scenarios with 82 people.

		No. Of people	No. Of people through main doors	No. Of people through emergency door doors	No. Of people through elevator	No. Of people through back doors	Computational runtime of PathFinder simulator (s)	Simulated total evacuation time (s)	Simulator time of 1st person evacuated (s)	Simulated time of last person evacuated (s)	Min travel distance (m)	Max travel distance (m)
1	Normal evacuation	82	82	0		0	3,8	105	4,1	104,8	4	98,4
2	normal plus emergency	82	39	43		0	3,4	94,3	4,1	94	4,7	99,5
3	Stairs at 9 m and above is blocked	82	0	82		0	4,1	110,8	4,7	110,6	5,6	131,8
4	Emergency door and back door	82	0	37		45	4,3	105,8	3,6	105,5	3,4	103
5	Back door closed at t = 14 s	82	0	70		12	4,9	129,3	3,5	129,1	3,2	130,8
6	Emergency Evacuation	82	0	82		0	6,9	110,5	4,7	110,5	4,7	95,4
7	Normal + elevator	82	82	0	21	0	6,4	161,3	4,1	161,2	4	84

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.autcon.2025.106225.

Data availability

Data will be made available on request.

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